

Deterministic differentiable exact-cone contact on GPUs: bit-identical adjoints, batched routing and system identification

Anton Björkegren

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Abstract

We describe a GPU contact solver that keeps the exact second-order Coulomb cone with the de Saxcé correction, returns analytical derivatives of the contact impulse and of the post-step velocity, and produces the same bytes on every run and on four GPU architectures. The step is a proximal ADMM with an exact per-contact cone projection and an outer Cadoux loop; the adjoint is the same matrix-free operator transposed, restricted by a per-contact active-set basis, solved by the same preconditioned CG. Cross-thread accumulation is int64 fixed point with a device-computed power-of-two scale, so the operator is order-independent.

Three measurements set the frame. First, the formulation matters at the millimetre: under the exact cone a sliding box holds a normal gap of $1.55\text{e-}11$ m, while the convex relaxation of the same code lifts it $4.35\text{e-}3$ m [build/RESULTS.md §B.c(ii)]. MuJoCo reaches 1.42 mm (default) and 4.37 mm (stiff) on the same scene parameters, in a rocking limit cycle with the contact count alternating 0/2/4 rather than in a stationary slide [build/U6/RESULTS.md §1(ii), §summary row 8]. Second, the iteration count falls where the cone binds: on the 1000:1 mass-ratio column PGS needs 18 225 sweeps against the proximal ADMM’s 216 [HUNT.md §Mätt/H11]. Wall-clock for that scene against PGS is not measured; the measured wall-clock crossover is $n_c = 4\,096$ ($1.55\times$) [build/U20/RESULTS.md §1]. On a 10 000-body lattice (40 000 contacts) ADMM reaches $2.4\text{e-}15$ in 25 iterations (84.09 ms) against PGS’s 1 000 sweeps to $7.3\text{e-}17$ (445.61 ms) in the same wall-clock step; PGS is $30\times$ over-converged, and at equal residual the interpolated PGS ladder is 860 sweeps and the ratio is $3.7\times$ against the h1 run (273/74.2 ms) and $3.25\times$ against the U20 step time (273/84.09 ms) [build/U20/RESULTS.md §1; build/h1/RESULTS.md §H1b.5]. Third, the derivatives are the ones a robotics stack already trusts: against Pinocchio’s analytical constrained-dynamics derivatives we match to $1.2\text{e-}13$ (Unitree A1), $1.3\text{e-}12$ (ANYmal C) and $1.8\text{e-}14$ (Talos) on dv^+/dt [build/h24/RESULTS.md §4, build/U19/RESULTS.md §1] — and then go past the bilateral formulation, whose force on a slipping foot sits 23 % outside the Coulomb cone [build/h24/RESULTS.md §9].

Determinism is measured, not asserted: bit-identical over repeated runs, and 26 of 26 hashed arrays byte-identical across L4, A10G and H100 [build/U20/RESULTS.md §2]. We name the two host-side causes that break cross-machine identity — the OpenBLAS microkernel family and nvrtc floating-point contraction — and the intervention that closes the first [build/digests/RESULTS.md §0b]. The price of deterministic accumulation is $1.10\text{--}1.16\times$ on the matvec after the scale update is fused [build/U3/RESULTS.md §summary]. The condition under which determinism is a requirement rather than a convenience is measured: under a non-contracting iteration budget on a 64-body, 256-contact scene, float32 atomics give 9 distinct active sets in 10 runs and a 113 % spread in $d\lambda/d\mu$, where int64 gives exactly 0 [build/h6/RESULTS.md §4].

Batched: 4096 all-stick stance scenes run in one captured CUDA graph at 17 047 env-steps/s and 120 393 adjoint directions/s on an L4, 0.9985 of them below $1\text{e-}8$ in 200 iterations, with env i alone byte-identical to env i inside the batch. On the cone-binding batch 77.7 % reach $1\text{e-}8$ in 2 000 iterations and the throughput at the 2 000-iteration accuracy point is not reported (h25 reports 9 559 env-steps/s and 68 488 env-directions/s at 200 iterations, where the solution is not

at the 77.7 % point) [build/U20/RESULTS.md §summary, §2.1; build/h25/RESULTS.md §Verdict row 4].

1 Introduction

1.1 The gap

Three properties are each available today, and no simulator we found carries all three at once: (a) an **exact** Coulomb cone with the de Saxcé correction, rather than a convex relaxation; (b) **analytical** derivatives of the contact solve, rather than finite differences or gradients of a penalty surrogate; (c) execution on a **GPU**, batched, and **bit-deterministic**.

- **Kamino** (Disney Research Zurich + NVIDIA, in Newton) solves the constrained multi-body forward dynamics as an NCP on the GPU, explicitly avoiding convexification, and demonstrates 4096 parallel environments; no differentiability claim was found [02_sota_ncp_mpm.md §A8; R1].
- **Simple** (Carpentier, Le Lidec and Montaut [R2]) casts compliant and rigid contact as an NCP solved with ADMM plus proximal regularization, and the companion work [R3] gives analytical derivatives of the whole pipeline by implicit differentiation — but it is a CPU, single-environment solver with no native batching or GPU path reported [02_sota_ncp_mpm.md §A2].
- **Dojo** [R4] solves a hard-contact NCP with a primal-dual interior-point method and differentiates a relaxed intermediate problem by the implicit function theorem; CPU/Julia, under a non-commercial research licence, and interior-point solvers are hard to warm-start across time steps [02_sota_ncp_mpm.md §A4, “Known hard problems” item 4]. No gradient of this work is compared against Dojo on any scene (§5.7).
- **MuJoCo / MJX / MJWarp** [R5, R6, R7] is the GPU-batched baseline, and its contact model is a convex relaxation of the friction cone. MuJoCo’s C API differentiates by finite differences (mjd_transitionFD); MJX is a JAX implementation and is differentiated by reverse-mode autodiff through the solver iterations. This work compares against the finite-difference route only [02_sota_ncp_mpm.md §A6; build/U6/RESULTS.md §2]; no measurement against MJX or Brax autodiff gradients exists in this campaign (§5.7).
- **Pinocchio** gives exact analytical derivatives of constrained forward dynamics — and is bilateral: the constraint is an equality on contact acceleration, the force is unconstrained in sign and has no cone [02_sota_ncp_mpm.md §A7; build/h24/RESULTS.md §1]. No paper titled “Pinocchio 3” exists; the derivatives rest on four works: the analytical rigid-body derivatives [R8a], the proximal constrained resolution [R8b], the constrained articulated body algorithms [R8c] and the library itself [R8d].
- **Differentiable Simulation of Hard Contacts with Soft Gradients for Learning and Control** [R9] (DiffMJX + CFD) repairs gradients of a *penalty* simulator; it is not an exact Signorini [R15] / Coulomb solver [02_sota_ncp_mpm.md §A9].

The negative claim is scoped: no GPU-native ADMM-for-NCP rigid-body solver beyond Kamino was found in the literature pass that produced this map, and that pass records it as a negative finding rather than a proven absence [02_sota_ncp_mpm.md §A10].

1.2 Why the exactness is not cosmetic

Dropping the de Saxcé term $\Gamma(\mathbf{u}) = (\mu\|\mathbf{u}_t\|, 0, 0)$ turns the NCP into the Anitescu convex relaxation [R10], and the relaxation’s slip branch admits $\mathbf{u}_n = \mu(\mathbf{u}_t \cdot \mathbf{d})$ instead of $\mathbf{u}_n = 0$ [build/RESULTS.md §A0]. A sliding box therefore hovers. The hover is not a bug in an implementation, it is the relaxation’s own prediction $\Delta t \cdot \mu \cdot \|\mathbf{u}_t\|$, and we measure it to within 4.4 % of that closed form [build/RESULTS.md §A4]. The same relaxation removes $7.93\times$ as much tangential momentum on the impacting step [build/RESULTS.md §A4].

1.3 Why determinism is a solver property, not an implementation detail

Non-associativity of float atomics is the standard obstruction to deterministic scatter of contact impulses onto shared bodies, and the standard deterministic alternatives are described in the literature as costing throughput [02_sota_ncp_mpm.md §“Known hard problems” item 3]. We treat determinism as a first-class requirement because a sensitivity computed on an active set that moves between runs is not a derivative of anything. The converse is measured and holds too: bit identity on a solve that has not converged is reproducibility, not accuracy — $d\lambda/d\tau$ is $4.569e-1$ from the CPU IFT reference on such an env [build/h25/RESULTS.md §4 item 3]. §3 gives the regime where that happens and the regime where it does not.

1.4 Contributions

1. A GPU proximal-ADMM contact step with an exact second-order cone projection and a device-side Cadoux/de Saxcé outer loop, matrix-free in $G = J M^{-1} J^T$, run entirely inside a captured CUDA graph (§2.2).
2. The adjoint as the same operator transposed: $Z^T(G+E)Z + D$, built from the same matvec and the same PCG, adding two block-diagonal 3×3 kernels and no new solver (§2.3).
3. int64 fixed-point accumulation with a device-computed power-of-two scale that preserves Krylov finite termination, plus the accumulate-the-total formulation that makes velocity a pure function of λ (§2.4).
4. Spectral ρ and η read from the pencil (L, M) of the contact graph without forming G , and a certificate-first router that reads the regime before the first ADMM iteration (§2.5, §2.6).
5. A determinism study that separates device causes from host causes by intervention, and states the measured condition under which determinism is required (§3).
6. Parity with Pinocchio’s analytical contact derivatives on three legged platforms, and a measured statement of where the bilateral model stops (§4.4).

2 Method

2.1 Formulation

Per contact c , components ordered $(n, t1, t2)$, rows contact-major:

$$u = G \lambda + b, \quad G = J M^{-1} J^T, \quad b = J v_{\text{free}} (+ \text{speculative offset})$$

$$\text{NCP: } K_{\mu} \ni \lambda_c \perp (u_c + \Gamma(u_c)) \in K_{\mu}^*, \quad \Gamma(u, \mu) = (\mu \|u_t\|, 0, 0), \quad K_{\mu}^* = K_{\{1/\mu\}}$$

The convex (Anitescu/MuJoCo-family) model is the same system with $\Gamma := 0$ [build/RESULTS.md §A0]. The per-contact local problem is solved exactly rather than by projection iteration: for a fixed slip direction d the slip branch is linear in λ_n , so it collapses to one scalar root find in the slip angle, and the exact cone costs what a projected update costs [build/RESULTS.md §A0].

2.2 Proximal ADMM with the exact cone and an outer Cadoux loop

Per step, with $E = \text{diag}(\eta_c I_3)$:

$$\min \frac{1}{2} \lambda^T (G+E) \lambda + (b + \Gamma(u))^T \lambda \quad \text{s.t.} \quad \lambda \in K_{\mu}$$

x-update	$(G + E + \rho I) x = -(b + \mu s + \gamma - \rho z)$	matrix-free PCG, block-Jacobi
z-update	$z = \text{proj}_{\{K_{\mu}\}}(x + \gamma/\rho)$	exact second-order cone
dual	$\gamma \leftarrow \gamma + \rho(x - z)$	
refresh	$u = (G+E)z + b; \quad s_c = \ u_{\{t,c\}}\ $	de Saxcé / Cadoux, on device

[build/h1/RESULTS.md §0]. **G is never formed.** One matvec is two sparse products per contact: k_{scat} accumulates $\Sigma_c M^{-1} J_c^T v_c$ per body in int64 fixed point, k_{acc2w} divides back, k_{gath}

reads $(J \ w)_c$ and adds $(\eta_c + \rho)v_c$ [build/h1/RESULTS.md §0].

The preconditioner is per-contact 3×3 block Jacobi, $P_c = (G_{cc} + (\eta_c + \rho)I)^{-1}$, built on device. A Laplacian sketch was measured and rejected: it buys **zero** outer iterations — identical counts to a dense Cholesky — and only changes the cost of the x-update [HUNT.md §Mätt/H11; build/h1/RESULTS.md §0].

The Cadoux refresh cadence matters and was tuned against two failure modes: refreshing every 25th iteration costs $5.5\times$, refreshing every iteration is a false attractor; the default is 4 [HUNT.md §Mätt/H11].

The proximal term is per contact, η_c , and it has a physical reading: η is contact compliance in 1/kg, and at $\eta = 0$ the ADMM's wrench is exact while λ itself is non-unique on hyperstatic scenes [HUNT.md §Mätt/H11]. The uniform-price law $\eta/(\eta + G_{cc})$ was **refuted**: the governing ratio is $\eta/\lambda_{\min}(G|_{\text{active}})$, measured at 1.75α (cube), $\approx 94\alpha$ (K8), $\geq 1669\alpha$ (1000:1 column) [build/h1/RESULTS.md §Verdict]. §2.5 gives the replacement.

2.3 The adjoint is the same PCG transposed

On the converged active set the residual system is, per contact: open $= \lambda_c = 0$; stick $= (G\lambda + b)_c = 0$; slip $= (G\lambda + b)_n = 0$ and $\lambda_t + \mu \lambda_n u_t / \|u_t\| = 0$ [build/RESULTS.md §A5]. Reducing by a per-contact 3×3 active-set basis Z gives the gradient operator

$$Z^T (G + E) Z + D$$

which is literally the same matrix-free `_matvec` pre- and post-multiplied by Z , solved by the same block-Jacobi PCG. Only two block-diagonal 3×3 kernels are added (`k_zmul`, `k_ztd`); no second solver exists [build/h1/RESULTS.md §Verdict].

The hypothesis “the KKT system is symmetric” is **partly** true and the missing piece is named: the de Saxcé term $\mu d\|u_t\|$ is rank-1 per slip contact and non-symmetric, because Coulomb friction is non-associated. It is absorbed by the *same* outer refresh the forward solve already runs — 1 refresh when nothing slips, ≤ 20 when everything does [build/h1/RESULTS.md §Verdict].

grad_affine. The σ -map of that refresh is affine, so the fixed point is solved exactly in a fixed small number of calls instead of iterated: 3 calls against 318, agreeing to $2.2e-11$ [HUNT.md §Mätt/H24]. As default it gives cube $4.7e-14$ and exactly 0 (one call) on column, tower and the A1 stance, byte-identically [HUNT.md §Mätt/U9]. Where the refresh contracts slowly the amplification is explicit: on the A1 slip probe $\rho(B) = 0.9183$, so a PCG residual error $\Delta B = 4.2e-12$ in B is amplified by $(1 - \rho(B))^{-2} = 149.8$ [build/U42/RESULTS.md §summary]. Two repairs were measured; a dense in-kernel LU on device for $n_c \leq 4$ (3 calls, 16.9 ms) closes it to $d\lambda/db = 4.54e-12$, 8–10 $\times$ faster than the Neumann defect correction [build/U42/RESULTS.md §summary].

Measured accuracy of the adjoint: $d\lambda/d\mu$ vs central FD $3.94e-8$ and vs the independent CPU IFT $1.7e-14$ on the pushed-cube patch; on the 1000:1 column and K8 tower $d\lambda/d\mu$ is exactly 0 by both routes, because every contact sticks and μ carries no information [build/h1/RESULTS.md §Verdict].

No comparison against reverse-mode autodiff of the unrolled ADMM iterations (`wp.Tape`) on the same scene exists; the adjoint is checked against central finite differences ($3.94e-8$), an independent CPU implicit-function reference ($1.7e-14$) and Pinocchio's analytical derivatives (§4.4) [build/h1/RESULTS.md §Verdict]. The solver is written in Warp [R12], which carries that tape, so the comparison is available and was not run (§5.7).

2.4 int64 fixed-point accumulation

Cross-thread accumulation uses `wp.atomic_add` on `wp.int64`. Integer addition is associative, so the sum does not depend on thread order [build/h1/RESULTS.md §0]. Every other reduction is order-

fixed at a compile-time width $NBLK = 256$, never at the launch configuration; there are no float atomics anywhere [build/h1/RESULTS.md §0; build/digests/RESULTS.md §0 candidate (d)]. The inventory is a source audit, not a build-time gate; no certificate row of §6.2 checks it [build/digests/RESULTS.md §0; build/cert/RESULTS.md §summary].

The scale cannot be a constant inside a Krylov method. With a fixed quantum $S = 1e10$ the CG residual eventually falls below the quantum and the matvec returns zero. The scale is therefore recomputed on device, per matvec, as a power of two:

$$S = 2^{\text{floor}(\log_2(2^{52} / (\kappa \cdot \max |x|)))}, \quad \kappa = \max_c \|M^{-1} J_c^T\| \text{ (host, once)}$$

Multiplication by an exact power of two is exact, and S is a pure function of x , so determinism survives. The consequence is that CG keeps finite termination: on K8 ($n = 96$) the gradient PCG residual is $6.2e-20$ at 96 iterations and $4.0e-43$ at 192 [build/h1/RESULTS.md §0]. The quantum sets the accuracy floor and the floor is linear in the bound: inflating κ by $1024\times$ inflates the error by $780\times$ [build/h24/RESULTS.md §8].

Accumulate the total, not the delta. An earlier formulation rounded each per-body velocity *delta* and left one rounding per contact update behind in v : a drift linear in the iteration count, ≈ 0.43 quantum per iteration. The accumulator now carries the per-body **total** $M^{-1} J^T \lambda$, and a contact update *replaces* its own quantized contribution by integer subtraction [build/RESULTS.md §B.b']:

iterations	before (float64)	after (float64)
500	2.139e-08	1.208e-10
80 000	4.365e-06	1.410e-10

a $31\,000\times$ improvement at 80 000 iterations, and flat in the iteration count [build/RESULTS.md §B.b']. This turns into a gate: because the accumulator is rebuilt exactly from the absolute λ , stopping and warm-restarting the solver is **bit-identical** to running straight through — 1×1200 iterations against 3×400 warm-restarted, λ bytes identical, v bytes identical [build/RESULTS.md §B.b']. It also runs faster at scale ($n_c = 4096$: $52.09 \rightarrow 40.21$ ms per 50 iterations) because the clearing pass over the body array is gone [build/RESULTS.md §B.b'].

2.5 Spectral ρ and η from the pencil

Diagonal rules for ρ and η fail by two orders of magnitude on mass-ratio problems: $\rho = \text{median}(\|G_{cc}\|_2)$ is within 15 % of best on the cube and **206 $\times$** worse than $\rho^*/10$ on the 1000:1 column (23 272 iterations against 113) [build/h1/RESULTS.md §1]. The reason is structural: the soft global mode of a 1000:1 stack is invisible to every contact diagonal [HUNT.md §Mätt/H11-H13 reconciliation].

The Delassus operator is, for the class where it holds, a grounded weighted Laplacian on the contact graph's line graph, and $\text{spec}(G)$ equals the spectrum of the generalized eigenproblem (L, M) — so the mass-ratio bottleneck is readable from the contact graph without forming G [HUNT.md §Mätt/H4]. Patch lumping with the 6×6 inverse spatial inertia as conductance makes the identity exact ($9.6e-17$) and survives rotation; what breaks it is body degree ≥ 3 [HUNT.md §Mätt/H13].

The estimator is matrix-free on the pencil $K^T K$ with $K = J \cdot M^{-1/2}$ [R18a], and gives $\lambda_{\min}^+/\lambda_{\max}$ to $1e-15 \dots 1.8e-9$ relative on all scenes including lattices, at 10-204 matvecs (≈ 6.4 matvecs per contact) [HUNT.md §Mätt/H15]. Ported to the GPU it needs one new kernel and costs 204 matvecs fixed in n_c , i.e. 2.5 % of the solve at $n_c = 4096$ [HUNT.md §Mätt/H1b].

Measured rules:

- $\rho = \lambda_{\min}^+$ beats $\sqrt{(\lambda_{\min} \lambda_{\max})}$: asymmetric tower 1 512 $\rightarrow$ 9 (168 $\times$), column at rest 213 $\rightarrow$ 36, K8 8 $\rightarrow$ 3, sliding column 216 $\rightarrow$ 213, cube 60 $\rightarrow$ 66 [HUNT.md §Mätt/H15]. On the GPU: K8 6 $\rightarrow$ 3, column at rest 67 $\rightarrow$ 35, sliding 83 $\rightarrow$ 52, asymmetric tower 278 $\rightarrow$ 9 [HUNT.md §Mätt/H1b].
- $\eta = \alpha \cdot \lambda_{\min}^+$ **bounds the wrench error by $\leq \alpha$** , $\text{err}/\alpha = 0.02\dots 1.00$ and constant in α , where the per-contact diagonal rule $\eta_c = \alpha \|G_{cc}\|_2$ gives 1.37...8 018 [HUNT.md §Mätt/H15].
- ρ also sits in the x-update, $\text{cond} = \lambda_{\max}/(\lambda_{\min}^+ + \eta + \rho)$: on the asymmetric tower 16 CG iterations diverge and 64 CG reproduce the CPU-exact 9 — the rule is “ $\lambda_{\min}^+$ with the x-step solved to the accuracy ρ demands” [HUNT.md §Mätt/H1b].
- η must be **frozen**, not re-estimated per Cadoux refresh: re-estimation costs 13.9 s against 0.29 s and is 46 % worse where the active set moves [HUNT.md §Mätt/H15].
- $\rho = \lambda_{\min}^+$ is **refuted on packings**: DEM packings want $\rho \approx 1.1\text{--}2 \cdot \sqrt{(\lambda_{\min} \lambda_{\max})}$, and no covariate separates the two families [HUNT.md §Mätt/H1b, §Mätt/H18].

2.6 Certificate-first router

Before any ADMM iteration, the **bilateral certificate** is evaluated: $\lambda_{\text{bil}} = -G^{-1}b$, and if λ_{bil} lies in the interior of K_{μ} it satisfies the NCP’s KKT conditions exactly, so the scene *is* all-stick — a certificate, not a probe. On the GPU it is a 12 $\times$ 12 in-kernel Cholesky, one thread per env, agreeing with NumPy on every env of both 4096-env probe batches [build/h25/RESULTS.md §“the two regimes are constructed”].

This replaces reading the regime from a trial solve, which was measured and **refuted**: an ADMM probe gives 37.1 % converged where the certificate gives 77.7 % for the same ρ rule — 2.1 $\times$ [build/h25/RESULTS.md §Verdict rows 4–5]. On the closed active set the certificate matches the slip-fraction covariate exactly *before* the first ADMM iteration [HUNT.md §Mätt/H23b].

The router is one entry point over 21 benchmark scenes drawn from the whole campaign [build/U18/RESULTS.md §summary]:

regime	route	measured
strictly interior	bilateral certificate, no ADMM	7/21 scenes in 1.0–6.3 sweep-equivalents
1-D mass-ratio chains	PGS + arc-freeze	43 sweeps against 18 225 plain PGS (423 $\times$)
granular packings	block-Ruiz [R17] + ADMM, $\rho = 0.1$	463–1 233 sweep-equivalents where PGS fails (>4 000 ... >20 000)
dense multi-body	ADMM ladder	246.8 and 2 198.4 sweep-equivalents
articulated, $n_c \leq 4$	PGS	27–53 sweeps where the ADMM ladder spends 209–485

U18’s per-scene “oracle” was a hard-coded source table, not an executed best solver per scene; on col8_cone the router cost was 0.106 $\times$ that table entry [build/A78/RESULTS.md rows 3, 54–60]. The replacement oracle is U135’s executed 25-point sweep per scene: ρ^* minimizes ADMM iterations over $\lambda_{\min}^+ \cdot \text{logspace}(-3, 3, 25)$, with cold starts, residual $< 1e-8$ and a cap of 5 000. The ratio below is router sweep-equivalents divided by iterations at ρ^* ; values below one can occur because the router also has certificate and PGS branches [build/A78/RESULTS.md §1; build/U135/RESULTS.md Table 6 v2].

scene	router sweep-equivalents	ρ^*	iterations at ρ^*	router / sweep oracle
cube	60.103501	0.1986272518	58	1.03626726 $\times$
K8	1.000000	2.333147240e-06	2	0.500000000 $\times$
col_rest	1.000000	2.927682520e-07	4	0.250000000 $\times$
col_slide	43.000000	0.0009258145028	162	0.265432099 $\times$
col_open	2.000000	0.001646356868	27	0.0740740741 $\times$

scene	router sweep-equivalents	ρ^*	iterations at ρ^*	router / sweep oracle
tower_asym	1.000000	3.778713140e-09	2	0.500000000×
lattice	6.345149	0.0004021119685	2	3.17257450×
rand256	246.849479	3.847139722	202	1.22202712×
dem_step	1233.344886	61.72496689	1104	1.11716022×
pack50	463.352946	493.5228954	577	0.803038035×
pack200	573.427699	487.0694009	833	0.688388594×
pack800	627.966164	477.3906827	926	0.678149205×
rand256_h21	2198.356655	225.3681962	800	2.74794582×
col_heavy_top	43.000000	0.01646356868	2681	0.0160387915×
col8_cone	300.000000	0.01352008815	3027	0.0991080278×
a1_stick	1.000000	1.683484815e-05	2	0.500000000×
a1_slip	485.000000	0.9466930820	627	0.773524721×
humanoid	1.000000	5.780543824e-05	3	0.333333333×
a1_h30	1.000000	2.721286926	56	0.0178571429×
h25_a1_stick	209.356840	0.1898863344	200	1.04678420×
h25_a1_slip	229.499483	0.5555643533	248	0.925401141×

Against PGS-100, the router uses $1.6637965955\times$ fewer sweeps and $3.1801101875\times$ less time in the 8/21 scenes where PGS-100 converges. Across all 21 scenes, including 13 PGS runs stopped unconverged at 100 sweeps, total time is 52.4367 s for the router and 35.7915 s for PGS-100 [build/A78/RESULTS.md rows 5–6, §2]. The router’s ρ is $1.6442790678\times$ from ρ^* on cube and $3.0871206049\times$ from ρ^* on a1_slip; three of the five audited scenes take a certificate or chain branch and receive no router ρ [build/A78/RESULTS.md rows 8–11, §4]. As a measured gate, the bilateral certificate misroutes 0/4 ANYmal C/Talos exports; this covers the four static exports, not A60’s time sequences [build/A78/RESULTS.md row 7, §3]. Adding the articulated branch takes the old table’s misrouting from 4/21 to 1/21 on the same 21 scenes, and the generalized $n_c \leq 4$ rule to 0/21 on the same 21 scenes; no leave-one-out error is reported for either, and both rules were selected on that set [build/U33/RESULTS.md §summary; HUNT.md §Mätt/U33].

2.7 Batched execution: 4096 envs in one CUDA graph

Every kernel carries the env index; the forward, the active-set build, grad_affine and the whole step+gradient are captured as one graph. The launch count is **133 per ADMM iteration regardless of B**, 35 436 graph nodes, one host launch per step+gradient, 13.8 kB per env [build/h25/RESULTS.md §Verdict row 2; HUNT.md §Mätt/H25]. Per env the choice is a dense 12×12 G with cyclic Jacobi, exact at $n_c = 4$, flipping to the matrix-free path at $n_c \geq 32$ [HUNT.md §Mätt/H25].

ρ has to be per env, not a batch scalar: $\lambda_{\max}$ is almost constant across the batch while $\lambda_{\min}^+$ spreads $3.6\times$ [build/h25/RESULTS.md §0].

3 Determinism

3.1 Bit identity within a machine

λ and v are byte-identical over two runs in float32 and float64, including with the contact list handed in shuffled (colouring is a pure function of a radix-sorted key) [build/RESULTS.md §B.a; HUNT.md §Mätt/B]. Gradient bytes are identical over two runs on every scene and every θ -direction [build/h1/RESULTS.md §Verdict]. Across the campaign: 21/21 router scenes byte-identical on CPU float64 [build/U18/RESULTS.md §summary], 21/21 on the GPU router

[build/U33/RESULTS.md §summary], full sha256 identical over two processes for the A1 stance controller’s 16 solver calls [build/h30/RESULTS.md §0].

The rule the campaign adopted after one lane claimed identity without one: **every bit-identity claim must carry a full sha256 of the full array** [HUNT.md §Mätt/Modal].

3.2 Bit identity across machines

comparison	result	source
L4 (sm_89) = A10G (sm_86) = H100 (sm_90), first pass	22/22 arrays byte-identical	HUNT.md §Mätt/Modal
the five suites that had no cloud identity	26/26 arrays, 5/5 suites byte-identical across all three	build/U20/RESULTS.md §2
4096-env batch, incl. 8 sampled envs, across L4/A10G/H100 and the local RTX 5070 (sm_120)	18/18 arrays identical; batch 4d5d4747... (stick), 24a370fe... (slip)	build/U20/RESULTS.md §2.1
local RTX 5070 vs cloud, device outputs	13 of 22 identical	build/RESULTS.md §B.i.3
local RTX 5070 vs cloud, host inputs	32 of 45 identical	build/RESULTS.md §B.i.3
the same comparison re-measured in a later lane	35 of 45 host, 16 of 22 device	build/A63/RESULTS.md §U56

No scene whose host inputs match has differing device outputs, over the 13 of 22 device outputs that matched in this run; the converse occurs (K8: host inputs differ, outputs identical), so the certificate bounds the cause from one side only. The counts move to 35/45 and 16/22 on re-measurement, so the statement is made for the set measured and not as a law [build/RESULTS.md §B.i.3; build/A63/RESULTS.md §U56].

3.3 The two causes, found by intervention

Four candidates were tested; two survived, and the one that mattered most was not on the candidate list [build/digests/RESULTS.md §0].

(a) nvrhc floating-point contraction — confirmed, for the lattice. With Warp’s module option `fuse_fp=False` an RTX 5070 (sm_120, nvcc 12.8.93) and a Modal L4 (sm_89, nvcc 12.9.86) reach the same 40-step state byte for byte; with the default `fuse_fp=True` they split at step 9 — 238 of 30 000 position components, ≤ 6 float32 ulp, $8.73e-11$ m [build/digests/RESULTS.md §0].

Contraction is a *compiler* choice: which multiply-adds get fused is a property of the nvrhc version and the target architecture, not of the kernel source. The solver therefore sets `fuse_fp=False` and `fast_math=False` as module options before the first kernel of the module is declared, with `WP_FUSE_FP=1` restoring the old build for A/B [build/RESULTS.md §B.i].

(e) the host — confirmed, for the 1000:1 column and for the ADMM solver. The column’s entire single-step iteration ladder (1...800 iterations) is byte-identical on the two devices under every codegen toggle; the first divergence in the 240-step run is at **step 3, in the host-built Ja, Jb, bias**, with λ , v , x , q and the residual still bit-identical [build/digests/RESULTS.md §0]. With no GPU in the picture, `np.dot` and `np.linalg.inv` give different bytes on the local AVX2 i5-13600K and the container’s AVX-512 CPU. The ADMM path is the same story through a different call: its host-computed ρ^* (`np.linalg.eigvalsh`) differs on cube and tower and is identical on the column — exactly the scenes that do and do not diverge [build/digests/RESULTS.md §0].

The intervention that settles it. `OPENBLAS_CORETYPE` selects the microkernel family inside a `DYNAMIC_ARCH` OpenBLAS [R14] at load time and changes nothing else. Running the container’s CPU probes three ways holds Python, NumPy, wheel and code fixed and moves only the kernel: `OPENBLAS_CORETYPE=Haswell` (what the local box selects) reproduces **every** local digest on the AVX-512 container — 13/13 and 21/21 — while the container default (SkylakeX) does not

[HUNT.md §Mätt/fuse_fp-fix; build/digests/RESULTS.md §0b]. Python 3.10 vs 3.12 is dead as a cause [HUNT.md §Mätt/fuse_fp-fix].

Two candidates were refuted: the device-computed power-of-two int64 scale (bit-identical over a 200 484-point sweep, including the 25 207 points where $\text{wp.pow}(2, e)$ is not exactly 2^e), and launch-config-dependent reduction order (complete inventory: every cross-thread accumulation is int64 or an integer counter/min/max; the only float reductions are fixed at compile-time width $\text{NBLK} = 256$) [build/digests/RESULTS.md §0]. float32-vs-float64 is refuted as the *discriminator*: float64 moves under contraction too; float32 sets the size of the amplification, not whether a perturbation exists [build/digests/RESULTS.md §0].

The counter-example worth keeping: the K8 tower's host Ja, Jb, bias and M^{-1} **differ** across the two CPUs at some step of 600, and its λ , v and trajectory are still bit-identical — a well-conditioned scene rounds the host difference away, the 1000:1 column amplifies it. Same cause, opposite visible outcome, which is why an output-only certificate cannot tell them apart and why the certificate hashes host inputs and device outputs separately [build/RESULTS.md §B.i.3].

One host gap is **not** closed by any in-Python fix: eigvalsh (dsyevd) leaves ρ^* differing by 111 ulp (cube), 33 ulp (tower), 0 (column) [HUNT.md §Mätt/fuse_fp-fix].

3.4 What determinism costs

measurement	value	source
matvec, int64 / float32 atomics, contended baseline	2.40–2.46×	build/h6/RESULTS.md §Verdict
one full ADMM iteration	1.38–1.55× (n_c 256), 1.53–2.33× (4096)	build/h6/RESULTS.md §Verdict
removable share of that price	~60 % (two reduction kernels for the device-side scale: 15.7 μs of a 39.9 μs int64 matvec, against a 16.6 μs float32 matvec)	build/h6/RESULTS.md §Verdict
matvec after fusing the scale update (1 per PCG solve instead of 1 per matvec), uncontended L4	1.103× (n_c 256), 1.161× (n_c 4096)	build/U3/RESULTS.md §summary
fused vs per-matvec scale, λ relative difference	1.7e-11 (256), 2.2e-11 (4096), 7.0e-14 (cube); gradient byte-identical, 4.8e-14 vs exact	build/U3/RESULTS.md §summary
scale updates per matvec after fusing	50/466 = 1 per 9.3	build/U3/RESULTS.md §summary
cost of fuse_fp=False	below this machine's noise floor, $ \text{ratio} - 1 \leq 2.5\%$ at n_c 256 and 4096	build/RESULTS.md §B.i.2

No alternative deterministic scatter (segmented sort-based reduction, per-body fixed-order reduction, compensated summation) was measured; the 1.103–1.161× is stated against float32 atomics only [build/U3/RESULTS.md §summary] (§5.7).

Switching contraction off flipped 2 of 43 gates — both threshold crossings on scenes this work already calls the sensitive ones (1000:1 drift 5.27e-3 against a 5e-3 gate; K8 jitter floor 8.80e-4 against 8.95e-4 m/s). The two gates stay in the suite and fail; the release criterion for a run that trips them is not defined in this work. What the flip measures is that those two gates were passing on a margin a compiler flag is enough to spend [build/RESULTS.md §B.i.1].

Determinism also buys accuracy, in the same measurement: at the same iteration budget and the same ρ , float32 atomics move the residual from 1.10e-11 to 4.81e-6 on the 1000:1 column, from 1.83e-11 to 7.47e-9 on the boundary cube, from 1.99e-12 to 2.36e-9 on K8 [build/h6/RESULTS.md §Verdict].

3.5 The measured condition under which determinism is required

int64 gives spread **exactly 0** in λ , in the active set, in $d\lambda/d\mu$, in $dv^+/d\mu$ and in the residual over 10 runs on all 10 scene/budget configurations measured [build/h6/RESULTS.md §Verdict].

float32 atomics lose bit-identity **only above one warp**. k_scat is one launch of n_c threads, and same-address atomics inside one warp are serialized by the hardware in lane order, so $n_c \leq 32$ is deterministic *for the wrong reason*; identity breaks at $n_c = 64$ [build/h6/RESULTS.md §2]. A contact solver that is deterministic on a 4-contact unit test and non-deterministic on a 64-contact scene is the default outcome, and the unit test never sees it.

The decisive scene is a 256-contact patch on one body with the tangential load sitting exactly at μN — maximum atomic contention, every contact on the stick/slip switch [build/h6/RESULTS.md §3]:

	int64	float32 atomics
λ spread, relative	0.000e+00	3.790e-06
runs off the majority active set	0/10 (1 distinct set)	0/10 (1 distinct set)
$d\lambda/d\mu$ relative spread	0.000e+00	8.650e-05
$dv^+/d\mu$ relative spread	0.000e+00	2.100e-04
residual min...max	8.725e-11 ... 8.725e-11	1.105e-08 ... 6.921e-08

Converged, the float32 column is $3.5e-6 \dots 8.7e-5$ — below HUNT’s $1e-3$ bar. **Under budget it is not, and it is measured on a second scene.** On a 256-contact synthetic scene of 64 bodies (syn256, not the patch16 scene of the table above) at $\rho = \rho$, *600 iterations and refresh cadence 8*, float32 atomics give **9 distinct active sets in 10 runs**, 8/10 runs off the majority, a residual wandering over $1.30e-2 \dots 3.33e-2$, and a **113 % relative spread in $d\lambda/d\mu$** (1.131 at cadence 8; $8.30e-2$ at cadence 25). int64 on the identical configuration gives one active set and 0. At $\rho = 3\rho$ the float32 spread is $2e-6$ at every budget [build/h6/RESULTS.md §4; HUNT.md §Mätt/H6].

The verdict, as measured: determinism is a *requirement* when the solver does not contract within budget — which no ρ rule yet predicts — and a *convenience* when it does. The solver cannot tell the caller which case it is in, which is why the requirement is taken as unconditional [build/h6/RESULTS.md §Verdict; HUNT.md §Mätt/H6]. The rule of §2.5 predicts the iteration count on chains, towers and lattices and is refuted on packings (§5.3) and at the near-open boundary (§5.2, 6 of 4096 envs diverge to $1e+55$); it does not predict whether a given budget contracts.

4 Results

4.1 The gliding artifact: what the exact cone buys, in millimetres

Scene (ii): box 0.20 m, 1 kg, $v_0 = 1$ m/s, $\mu = 0.3$, $dt = 1/240$ s (CPU reference: $dt = 0.004$ s), 200 steps, 4 contacts, no position stabilisation.

engine / formulation	max normal gap during the slide	stopping distance vs $v_0^2/2\mu g = 0.169895$ m
ours, exact cone + de Saxcé (GPU)	1.552204e-11 m [build/RESULTS.md §B.c(ii)]	0.167817 m (−1.223 %) [build/U6/RESULTS.md §1(ii)]
ours, convex relaxation (same code, desaxce=False)	4.352523e-03 m [build/RESULTS.md §B.c(ii)]	0.167815 m (−1.224 %) [build/U6/RESULTS.md §1(ii)]
ours, CPU reference, exact cone	1.092e-14 m [build/RESULTS.md §A4]	—
ours, CPU reference, Anitescu	4.320e-03 m peak, 5.020e-04 m mean [build/RESULTS.md §A4]	—

engine / formulation	max normal gap during the slide	stopping distance vs $v_0^2/2\mu g = 0.169895$ m
MuJoCo 3.9.0, default contact	1.42e-03 m [build/U6/RESULTS.md §1(ii)]	0.168087 m (−1.064 %)
MuJoCo 3.9.0, stiff	4.37e-03 m [build/U6/RESULTS.md §1(ii)]	0.167858 m (−1.199 %)
Box3D [R11] (external CPU oracle)	−8.07e-06 m [build/U6/RESULTS.md §1(ii)]	0.169374 m (−0.306 %)

The two MuJoCo rows are not a stationary slide: contacts alternate 0/2/4 and the box rocks, so their gap column is a limit-cycle amplitude and not a hover height [build/U6/RESULTS.md §summary row 8].

The measured hover matches Anitescu’s own prediction $\Delta t \cdot \mu \cdot \|u_t\|$ to within 4.4 % on the mean (ratio 1.0442, min 1.0156, max 1.1985; the maximum is the first-impact transient) [build/RESULTS.md §A4]. The second consequence of dropping Γ is on the impacting step: the relaxation removes **7.93×** as much tangential momentum, NCP 1.1772e-02 m/s (exact: $\mu g \Delta t$) against CCP 9.3369e-02 m/s [build/RESULTS.md §A4].

MuJoCo’s convex model differs from the exact one in the forward trajectory too: on the pushed cube it travels 82 mm further in x (0.4955 m against 0.4138 m) and lifts 6–16 mm where the exact solution stands on the plane [build/U6/RESULTS.md §1]. Box3D’s stopping distance is closer to the closed form than ours (−0.31 % against −1.22 %) — and the gap is **substepping, not the solver**: Box3D with 1 substep gives −1.26 % [HUNT.md §Mätt/Box3D-oraklet], and our own solver hits −0.31 % exactly when run at $S = 4$ substeps, independently of iterations and formulation [HUNT.md §Mätt/B3].

The adversarial search of §4.6 puts a number on the relaxation’s cost across a scene family: the convex solver’s hover reaches $\text{pen}/R = 7.27$ on stacked-box scenes [build/U23/RESULTS.md §summary].

4.2 Iteration counts where the cone binds

scene	PGS	proximal ADMM	source
1000:1 column, cone binding	18 225	216	HUNT.md §Mätt/H11
8-contact 1000:1	> 30 000	2 826	HUNT.md §Mätt/H11
asymmetric 1/1000 tower	nothing converged	1 512	HUNT.md §Mätt/H11
K8 tower	14 682	8	HUNT.md §Mätt/H11
1000:1, frictionless	28 456	1	HUNT.md §Mätt/H13
pushed cube	20	60 (a loss)	HUNT.md §Mätt/H11

The value sits in the stiff, many-contact regime; on the well-conditioned cube PGS wins. With the spectral ρ of §2.5 the same counts drop further: asymmetric tower 1 512 $\rightarrow$ 9, K8 8 $\rightarrow$ 3 [HUNT.md §Mätt/H15]. The 216 was cross-checked against an independent lane that first reported > 30 000 on a related scene; the discrepancy resolved to scene, η and residual definition, and the 216 reproduces exactly [HUNT.md §Mätt/H11-H13 reconciliation].

Per-iteration cost is not per-sweep cost: one ADMM iteration is 0.09× / 0.82× one PGS sweep at $n_c = 32 / 256$, because PGS needs n_c scalar root finds while ADMM needs one closed-form projection [HUNT.md §Mätt/H11].

4.3 Wall-clock, uncontended

All rows on a dedicated Modal L4 with 0.04–0.7 % repeat spread [build/U20/RESULTS.md §summary]:

measurement	PGS	ADMM	ratio	source
n_c = 32	40.45 ms (200 sweeps)	93.39 ms (200 it)	0.43×	build/U20/RESULTS.md §1
n_c = 256	72.58 ms (200 sweeps)	188.86 ms (275 it)	0.38×	build/U20/RESULTS.md §1, §3.1
n_c = 4 096	1 139.29 ms (1 000 sweeps)	735.70 ms (825 it)	1.55×	build/U20/RESULTS.md §1
N = 10 000 lattice, 40 000 contacts	445.61 ms (1 000 sweeps)	84.09 ms (25 it, $\rho = \lambda_{\min}^+$)	5.30× (same wall-clock step)	build/U20/RESULTS.md §1, §summary

The lattice row is not an equal-residual comparison: PGS stops at $7.3\text{e-}17$, ADMM at $2.4\text{e-}15$, so PGS is $30\times$ over-converged; interpolating the PGS ladder puts equal-residual PGS at 860 sweeps ≈ 273 ms and the ratio at $3.7\times$ against the h1 run ($273/74.2$ ms), $3.25\times$ against the U20 step time of 84.09 ms [build/h1/RESULTS.md §H1b.5; build/A75/RESULTS.md]. Both numbers are stated; $5.30\times$ is the ratio of the two wall-clock steps as run.

An archived claim of $3.47\times$ at $n_c = 256$ is **refuted** by these numbers: locally PGS landed at $1.632\text{e-}8$ at 200 sweeps, just over the $1\text{e-}8$ gate, forcing the geometric ladder up to 1 000 sweeps (684 ms); uncontended on the L4 PGS stops at 200 sweeps with $2.83\text{e-}9$. The archived figure was a sweep-step artefact [build/U20/RESULTS.md §3.1]. ADMM wins from $n_c \geq 4\,096$ on synthetic scenes, and on the lattice by $5.30\times$ per step as run and $3.25\text{--}3.7\times$ at equal residual (U20 step time or h1 run).

4.4 Parity with Pinocchio’s analytical contact derivatives

The correspondence is exact and integrator-free. Choosing the stance at rest ($v = 0$) makes $\dot{v} = \dot{v} = 0$, and with $v_{\text{free}} = v + dt\,M^{-1}(\tau - b)$ the acceleration-level and impulse-level systems are the same system scaled by dt : $\lambda = dt \cdot f$, $v^+ = dt \cdot a$, $d\lambda/d\tau = dt \cdot d\lambda_{\text{pin}}/d\tau$, $dv^+/dq = dt \cdot ddq_{\text{dq}}$ [build/h24/RESULTS.md §1]. That is why the errors below are $1\text{e-}13$ and not $1\text{e-}6$.

The GPU side is h1’s solver with one adaptation: a dense $nv \times nv$ M^{-1} matvec replacing the block-diagonal one (four new Warp kernels for the matvec, two for the preconditioner diagonal). The exact cone projection, ADMM updates, device-side Cadoux refresh, block-Jacobi PCG, active-set build and gradient solve are unchanged, and the int64 determinism pattern is untouched [build/h24/RESULTS.md §2].

quantity	Unitree A1 (nv 18, n_c 4)	ANYmal C (nv 18, n_c 4)	Talos (nv 38, n_c 2)	MuJoCo humanoid (nv 27, n_c 2)
rel err λ	3.58e-11	1.89e-11	4.82e-11	4.70e-11
rel err v^+	1.28e-10	1.95e-11	4.84e-11	4.34e-11
$dv^+/d\tau$	1.22e-13	1.27e-12	1.75e-14	2.94e-13
$d\lambda/d\tau$	3.86e-12	3.51e-13	7.20e-12	1.51e-12
dv^+/dq	2.99e-11	7.56e-10	1.09e-09 (all 38 DOF)	1.58e-09 on 11 of 27 DOF
$d\lambda/dq$	1.62e-10	4.75e-10	5.23e-10 (all 38 DOF)	1.02e-09 on 11 of 27 DOF

Sources: [build/h24/RESULTS.md §3, §4] for A1 and humanoid, [build/U19/RESULTS.md §1] for ANYmal C and Talos. The q-derivatives come from our adjoint **composed** with Pinocchio’s kinematics (dJ/dq , dM^{-1}/dq , dv_{free}/dq); no finite difference passes through our solver [build/h24/RESULTS.md §4]. The sign convention is settled rather than fitted: the flipped comparison gives exactly 2.000000 for every entry [build/h24/RESULTS.md §4].

Two rows must be quoted with their qualification. **The humanoid d/dq row is 11 of 27 DOF**, because Pinocchio 4.0.0’s own analytical q-derivatives are wrong on JointModelComposite joints: a contact-free probe comparing `computeABADerivatives().ddq_dq` against central FD of `pin.aba` —

Pinocchio against Pinocchio — gives an h-independent 41 % error concentrated exactly on the composite DOFs [build/h24/RESULTS.md §6]. Talos has 0 composite DOFs and needs no restriction [build/U19/RESULTS.md §1]. **dv⁺/dv is a different derivative by construction:** Pinocchio’s constraint is $J a = 0$, ours is $J v^+ = 0$, and those coincide only at $J v = 0$; the two differ by 5.25 (ANYmal C), 5.52 (Talos), 6.83 (A1), 12.7 (humanoid) [build/h24/RESULTS.md §7; build/U19/RESULTS.md §1].

Where the bilateral model stops. Same A1 stance, one extra torque (3 N·m on the rear-left knee), $\mu = 0.5$, chosen so exactly one foot leaves the cone while every λ_n stays positive. Our labels match the CPU reference PGS exactly ([stick, stick, slip, stick], $|\lambda_{\text{gpu}} - \lambda_{\text{cpu}}|_{\text{rel}} = 4.56\text{e-}08$). The cone check is the point: $\|\lambda_t\| - \mu\lambda_n$ is $[-6.48\text{e-}2, -2.70\text{e-}2, 0.0, -6.20\text{e-}2]$ for ours — the slipping foot sits *on* the cone at $\|u_t\| = 0.1254$ m/s — and $[-5.79\text{e-}2, -3.41\text{e-}2, +8.82\text{e-}3, -5.47\text{e-}2]$ for Pinocchio. **His rear-left force is 23 % beyond what that foot can carry**, because a bilateral constraint has no cone to be outside of; load then redistributes and the slipping foot sheds 38 % of its normal impulse onto the diagonal pair [build/h24/RESULTS.md §9]. On ANYmal C the same construction gives a cone violation of $+1.08\text{e-}3$ on the slipping foot against our 0.00 [build/U19/RESULTS.md §2]. And $d\lambda/d\mu$ — a derivative his tool does not have — agrees with the CPU IFT to $2.21\text{e-}11$ (A1) [build/h24/RESULTS.md §9], $2.41\text{e-}13$ (ANYmal C) and $2.31\text{e-}11$ (Talos) [build/U19/RESULTS.md §summary].

All of it is bit-identical, with full sha256 over two process invocations, on every platform [build/h24/RESULTS.md §5; build/U19/RESULTS.md §1].

One row of this table was wrong, and the correction is reported here. An audit of the batched `grad_affine` on a four-contact ANYmal C push scene where all four contacts slip put the GPU adjoint 0.7923098820556508 in maximum absolute error from the CPU IFT, relative Frobenius error 0.1646595489506769 — with the active set matching contact for contact in 8 of 8 samples, a converged NCP (natural residual $4.440892098500626\text{e-}16$), and one-sided finite differences at $h = 1\text{e-}6$ bracketing the CPU value, so neither a moving active set nor the cone edge accounts for it; 60, 120 and 240 gradient-PCG steps give the same error [build/A71/RESULTS.md]. The cause is one missing half of a guard: `k_setsig(h25_gpu.py:648)` tested `j < nslip[e]` and not `j >= 0`, while `grad_affine` calls the same kernel with the sentinel `j = -1` for its “ $\sigma = 0$ ” evaluation. Where the preceding env has $n_{\text{slip}} = n_c$, that read lands on a real slip slot, so $a = S(0)$ is evaluated at $\sigma = e_{\{C-1\}}$, every column difference $S(e_j) - a$ collapses to 0, $B = 0$, and the affine fixed point of §2.3 drops out [build/U110/RESULTS.md §2]. After the guard is completed, the same task gives $1.0658141036401503\text{e-}13$ maximum absolute error and $2.681134391940748\text{e-}14$ relative Frobenius error, and A1 ($7.998046669399628\text{e-}12$) and Talos ($2.1316282072803006\text{e-}14$) are unchanged to the byte, same array sha256 [build/U110/RESULTS.md §4].

A later measurement widens that scope. The sentinel write lands on `sig[e·C + sidx[e·C - 1]]`, contact $C-1$, in every case, and corrupts when and only when that contact slips, because `k_grhs` and `k_sigma` read σ only where `slipf > 0.5`. Two paths hit: where the preceding env has $n_{\text{slip}} = n_c$, `sidx[e·C - 1] = C-1` and the write lands on contact $C-1$ of the env itself; where the preceding env has $n_{\text{slip}} < n_c$, `sidx[e·C - 1] = -1` is the fill value of `k_slipidx` and the write lands on contact $C-1$ of the preceding env. The unfixed `grad_affine` is therefore wrong for every batch in which contact $C-1$ of any env slips; $n_{\text{slip}} = n_c$ is sufficient, not necessary. The chain scene `partial_4` ($n_c = 4$, $n_{\text{slip}} = 3$, the sticking contact first) falls on the unfixed copy at $3.166\text{e-}02$ in $d\lambda/db$ and $4.576\text{e-}02$ in $d\lambda/d\mu$, while the same scene with the sticking contact last (`partial_4_last`) is bit-identical between the two copies [build/U114/RESULTS.md §3]. The defect moves neither the residual, nor the active set, nor any determinism digest — it shows only against an independent reference [build/U110/RESULTS.md §7; build/U114/RESULTS.md §3].

Nine scenes carry the exactness rows, including chain scenes (contact c couples body c to body $c+1$, the last to the ground) in which every contact slips, and two in which one contact sticks. Each scene is run through the batched `grad_affine` against the CPU IFT (`ncp_ref.sensitivity`) with the unfixed and the fixed kernel copy; both copies receive bit-identical G , η , λ and reference matrices, so the kernel is the only variable [build/U114/RESULTS.md §1]. `cube`, `column` and `tower` are evaluated on $G + E$ with $\alpha_\eta = 1e-2$, because their raw Delassus operator is rank-deficient (cube rank 6/12, tower rank 48/96); the chain scenes are evaluated at $\alpha_\eta = 0$. `column`'s $3.401e-13$ is the floor of the 1 500-iteration PGS reference, not a kernel error [build/U114/RESULTS.md §1].

scene	n_c	n_slip	labels	max rel err $d\lambda/db$, unfixed	max rel err $d\lambda/d\mu$, unfixed	max rel err $d\lambda/db$, fixed	max rel err $d\lambda/d\mu$, fixed	bit-identical over two runs
cube	4	4	slip=4	2.179e-01	9.920e-01	1.326e-13	1.427e-13	yes
column	3	0	stick=3	3.401e-13	0.000e+00	3.401e-13	0.000e+00	yes
tower	32	0	stick=32	1.751e-09	0.000e+00	1.751e-09	0.000e+00	yes
all_slip_1	1	1	slip=1	3.610e-02	5.758e-02	2.220e-16	1.406e-16	yes
all_slip_2	2	2	slip=2	3.498e-02	5.413e-02	1.243e-15	2.044e-16	yes
all_slip_4	4	4	slip=4	3.315e-02	5.121e-02	8.120e-15	1.548e-14	yes
all_slip_8	8	8	slip=8	3.020e-02	5.018e-02	6.532e-14	2.401e-12	yes
partial_4	4	3	slip=3;stick=1	3.166e-02	4.576e-02	5.329e-15	1.747e-14	yes
partial_4_last	4	3	slip=3;stick=1	1.104e-14	2.044e-14	1.104e-14	2.044e-14	yes

[build/U114/RESULTS.md §2; build/U114/U114_table.md]. The fixed copy stays at or below $1.751285e-09$ in $d\lambda/db$ and $2.400576e-12$ in $d\lambda/d\mu$ over all nine scenes. The unfixed copy is wrong on six of them and bit-for-bit identical with the fixed copy on the three where no contact slips — `column`, `tower` and `partial_4_last`, sha256 of $d\lambda/db$ equal in both copies — which bounds the defect to the slip path [build/U114/RESULTS.md §2, §3]. `tower` at $1.751285e-09$ is PCG truncation and not a kernel error: it drops to $2.398170e-12$ at $cg = 120$ and does not move to $cg = 480$, while every other row is unchanged over that sweep [build/U114/RESULTS.md §4]. Two separate processes per copy give identical sha256 for all 18 gradient matrices, and two graph replays in one process give the same in all 36 cases [build/U114/RESULTS.md §5].

`cube` separates the two code paths. Through the batched path the unfixed copy gives $2.179e-01$ in $d\lambda/db$ and $9.920e-01$ in $d\lambda/d\mu$ on that scene; `h1`'s own `t2_gradient` row for the same scene stays green because it calls the host-side `grad_affine`, which passes a real zero vector. The sentinel encoding exists only in the batched path, and no earlier row reached it [build/U114/RESULTS.md §summary; build/U110/RESULTS.md §5].

4.5 4096 environments in one launch, with the adjoint

Unitree A1, $nv = 18$, 4 toe contacts, $dt = 1/240$, $B = 4096$ independently perturbed stance scenes, float64.

measurement	value	source
env i alone == env i in the batch	8/8 envs, full sha256 over $(\lambda, v^+, d\lambda/d\tau, dv^+/d\tau, d\lambda/d\mu)$, both batches	build/h25/RESULTS.md §1
— including on a non-converged batch	slip batch, 1/4096 below $1e-8$: mid-iteration ADMM state, duals and a moving active set, still 8/8	build/h25/RESULTS.md §1

measurement	value	source
— and on such an env the adjoint is wrong	the B = 64 slip probe gives $d\lambda/d\tau$ 4.569e-1 against the CPU IFT at a residual of $\sim 1e-3$ while remaining bit-identical alone and in the batch; bit identity is a claim about reproducibility, never about accuracy	build/h25/RESULTS.md §1, §4 item 3
— across L4, A10G, H100 and local sm_120	18/18 arrays identical	build/U20/RESULTS.md §2.1
throughput vs B = 1	$\times 3.74$ wall time for $\times 4096$ work = 1096$\times$	build/h25/RESULTS.md §Verdict
forward, L4 uncontended, all-stick batch	17 047 env-steps/s (240.28 ms), 0.9985 of the batch below $1e-8$ in 200 iterations	build/U20/RESULTS.md §summary
adjoint, L4 uncontended, all-stick batch	120 393 env-directions/s (68.04 ms)	build/U20/RESULTS.md §summary
the cone-binding batch	77.7 % below $1e-8$ at 2 000 iterations, q90 $1.33e-7$, max $1.09e-4$; no throughput is reported for this regime	build/h25/RESULTS.md §Verdict row 4, §4 item 2
the same two on the L4 timing table (separate repetition)	16 803 env-steps/s (243.76 ms); 118 972 dir/s (68.86 ms)	build/U20/RESULTS.md §1
full 18-column $dv^+/d\tau$	2 254 env-Jacobians/s	HUNT.md §Mätt/H25
adjoint accuracy per env vs CPU IFT	$d\lambda/d\tau$ $4.5e-13 \dots 3.4e-11$, $dv^+/d\tau$ $1.8e-12 \dots 3.1e-11$, $d\lambda/d\mu$ $6.8e-12 \dots 6.9e-11$	build/h25/RESULTS.md §Verdict
$\rho = \lambda_{\min}^+$ on 4096 all-stick envs	99.85 % below $1e-8$ in 50 iterations, where $\sqrt{(\lambda_{\min} \lambda_{\max})}$ needs 400 ($8\times$) and $\lambda_{\max}$ never gets there	build/h25/RESULTS.md §Verdict

Every throughput number in this table is measured on an all-stick batch, which is the regime the certificate of §2.6 routes *past* ADMM; no throughput for a cone-binding batch is measured anywhere in this campaign (§5.7) [build/h25/RESULTS.md §Verdict rows 4–5].

On ANYmal C the same batch gives 17 561 env-steps/s, 1078 $\times$ scaling and 82 868 adjoint directions/s, with 8/8 env-alone == in-batch [build/U19/RESULTS.md §summary].

One caveat is stated rather than hidden: the number of `grad_affine` calls is `n_slip_max + 2` taken over the batch, i.e. a graph-shape parameter, so an env re-run alone must be given the same call count (the extra calls run with $\sigma = 0$ and their columns are ignored, so the arithmetic is unchanged). Forward-only identity has no such caveat [build/h25/RESULTS.md §1].

4.6 Slip probing as a stance controller

A quasi-static torque-level probe unloads one A1 foot, induces micro-slip, reads λ and v^+ , and updates a per-foot μ estimate from the Fisher information of our adjoint.

claim	measured	source
exactly one foot slips per probe	A1: certificate confirms 7/8 probes (seed 12), 8/8 (seed 5), 7/8 (seed 31). ANYmal C: 6/8, 6/8, 5/8	build/h30/RESULTS.md §0; build/U19/RESULTS.md §summary
a stuck contact carries no information about its μ	$\ d\lambda/d\mu_c\ = 0.000000e+0$ bit-exact for every stuck foot; $\text{rank}(F) = \text{number of feet that have slipped, at every } k$	build/h30/RESULTS.md §0
four μ estimated after 8 probes	$\max \hat{\mu} - \mu = 7.85e-4$, max CRB 9.71e-4 (seed 12); $1.17e-3 / 9.73e-4$ (seed 5); $1.03e-3 / 9.77e-4$ (seed 31)	build/h30/RESULTS.md §0

claim	measured	source
on ANYmal C	max CRB $1.12\text{e-}3$ / $1.14\text{e-}3$ / $1.18\text{e-}3$ over three seeds, rank 4/4 at $k = 6\dots 8$, $\Sigma \tau = 358.0$ N·m, max $ \tau _\infty = 27.2$ N·m against an 80 N·m limit	build/U19/RESULTS.md §summary
body-level push baseline	best rank(F) over 36 pushes ($0.02 \rightarrow 100$ N·s, x/y/yaw): 2/4 ; union over all 36: 2/4; $\sigma_{\min} = 0$ and CRB = ∞ at every magnitude	build/h30/RESULTS.md §0
why	99.96 % of a horizontal base impulse becomes trunk momentum, not contact impulse	build/h30/RESULTS.md §1.2
on ANYmal C, body pushes	rank 0/4 below 6 N·s; rank 4 only at 15–40 N·s, where all four feet slip and the stance collapses	build/U19/RESULTS.md §summary

Two of the hypothesis’s own prescriptions were refuted by the measurement and replaced. A margin of “tangential impulse just above $\hat{\mu}\lambda_n$ ” (5 %) leaves a tangential deficit of $1.05\text{e-}3$ N·s $\approx \sigma_f$ and a slip speed of $5.8\text{e-}3$ m/s below the 5σ floor of $1.06\text{e-}2$ m/s: a real slip is invisible to both observables, the controller certifies a false lower bound, and the error jumps from $4.9\text{e-}5$ to 0.160. The repair is a measured floor, $m \cdot \hat{\mu} \cdot f \cdot \lambda_{\{n, \text{nom}\}} \geq 8\sigma_f$ [build/h30/RESULTS.md §0]. And the circle rule that picks the next foot holds **as a coverage rule** ($k = 1..4$ covers all four feet, cheapest first) and fails after coverage, where every belief saturates and plain round-robin beats it $1.73\times$ on $\sigma_{\min}$ for 1.5 % less torque — a rank tiebreak on the plateau is needed [build/h30/RESULTS.md §0; HUNT.md §Mätt/H30].

The earlier obstruction (“four loaded feet make the sensitivity undefined”) does **not** transfer to an articulated robot: it was a rigid-trunk artefact. A1 has rank(J) = 12/12, cond(G) = $3.3181\text{e}2$, and the adjoint agrees with the CPU IFT to $1.83\text{e-}8$ with all four feet loaded [build/h30/RESULTS.md §0].

4.7 Adversarial scene search

448 scenes were simulated through five solver channels (Box3D, MuJoCo, our exact GPU, our convex GPU, our CPU reference) in a 6-D scene space: $N \in [2, 12]$ boxes, mass ratio $r \in [1, 1000]$, $\mu \in [0.1, 1.0]$, $dt \in [1/960, 1/120]$, side load $s \in [0, 2] \cdot \mu\text{N}$, box proportion $P \in [0.5, 2]$ [build/U23/RESULTS.md §1].

search	median disagreement	source
random baseline (64 scenes)	16.34	build/U23/RESULTS.md §3
random rounds R1–R4	8.66 / 15.93 / 19.87 / 17.15 (mean 15.40)	build/U23/RESULTS.md §3
DPP throws weighted by the disagreement field	29.64 / 33.22 / 33.78 / 39.15 (2.40× baseline, 2.28× random)	build/U23/RESULTS.md §3
gradient-free local search	38.30 / 47.53 / 48.83 / 54.15 (exploits peaks; DPP diversifies)	build/U23/RESULTS.md §3

Disagreement is largest at tall stacks ($N \geq 6$) with mass ratio > 100 and a side load near the tipping/sliding boundary [build/U23/RESULTS.md §summary].

The honest finding is a loss: the `gpu_exact` channel — the exact cone projection driven by a fixed 100-sweep GPU PGS, not the ADMM step of §2.2 — **loses against the analytic anchor in 39 of 448 scenes**, e.g. scene #1 ($N = 11$, $r = 427.2$, $\mu = 0.93$) where it sinks 357 mm (pen/R = 7.14) against Box3D’s 35 mm (pen/R = 0.71) [build/U23/RESULTS.md §summary, §4]. The attribution U23 attached to that loss — Delassus ill-conditioning in the PGS path — is **not supported**: raising the sweep cap tenfold on the five worst scenes moves pen/R by less than 0.4, and four

solvers that all converge land at 5.026 / 5.881 / 6.500 / 7.136 against Box3D’s 0.706 on scene #1 [build/P2close/RESULTS.md §A.3, §A.5; §4.9].

Four repairs were measured on the same 39 scenes against a PGS-100 control, under U23’s own loss rule, with the 39 control wins re-run as the regression check where the lane reports it. Every channel in this table, including the control and the gpu_exact baseline, truncates the velocity solve at a fixed sweep count; “exact” refers to the cone projection and the de Saxcé term, not to the convergence of the iteration [build/P2close/RESULTS.md §A.2, §A.3]. The contact-path column names which contact code produced the row, because the U44b contact cache was defective until the fix reported below:

repair	repaired / 39	regressions / 39 wins	contact path	source
plain PGS-100 control, CPU world	5	—	U23 harness (fast_contacts), no U44b cache	build/P2close/RESULT S.md §A.3
U18/U33 router instead of PGS-100	6	—	U23 harness (fast_contacts), no U44b cache	build/P2close/RESULT S.md §A.3
fixed substepping S = 2 / 4 / 8	8 / 9 / 10	—	U23 harness; the lane reports no U44b cache	build/U76/RESULTS.m d §4
adaptive substepping, criterion A	10 (mean n_sub 3.42)	2	U23 harness; the lane reports no U44b cache	build/U76/RESULTS.m d §4
adaptive substepping, criterion B	9 (mean n_sub 3.19)	0	U23 harness; the lane reports no U44b cache	build/U76/RESULTS.m d §4
NGS position projection, S = 1	6	—	old (defective) U44b cache	build/U81/RESULTS.m d §summary
NGS position projection, S = 4	12	—	old (defective) U44b cache	build/U81/RESULTS.m d §summary
ADMM (spectral ρ , device Cholesky) + NGS, S = 4	8	1	old (defective) U44b cache	build/U97/RESULTS.m d §8
the same ADMM at S = 1	2	—	old (defective) U44b cache	build/U97/RESULTS.m d §8
the same ADMM at S = 4 without NGS	6	—	old (defective) U44b cache	build/U97/RESULTS.m d §8
PGS re-run under the fixed contact path	6	1	fixed contact path (SAT, no U44b cache)	build/U113/RESULTS. md §“Torn och U23”
ADMM re-run under the fixed contact path	7	1	fixed contact path (SAT, no U44b cache)	build/U113/RESULTS. md §“Torn och U23”

The four bold rows are the ones measured in the configuration this work recommends. The 12 of 39 is the highest number in the table and it was produced under the defective cache; it is not re-run, and nothing in this paper is closed on it.

Criterion A is $K = \max_c |v_{\{n,c\}}| \cdot dt / \max(\text{gap}_c, \varepsilon)$ with $\varepsilon = 0.1$ mm and $n_{\text{sub}} = \text{clamp}([K], 1, 8)$; criterion B is $n_{\text{sub}} = 1 +$ the number of contacts entering or leaving the active set on the previous step [build/U76/RESULTS.md §3]. NGS is run as a direct position update at fixed velocity: $\beta \cdot C / dt_{\text{sub}}$ as the position right-hand side, $\mu = 0$, slop 0.5 mm, 8 sweeps over the coloured contact sets, and the offset applied to x and q while v and ω are left untouched [build/U81/RESULTS.md §2].

Three statements follow from the ladder. First, substepping alone tops out at 10 of 39, at mean n_{sub} 3.42 and 2 regressions on the wins; criterion B holds 0 regressions and repairs 9 [build/U76/RESULTS.md §4, §7]. Second, on scenes with a side load $s > 1$ substepping raises the

slip error rather than lowering it: scene #2's final lateral slip runs 297.2 mm ($S = 1$), 610.7 ($S = 2$), 807.5 ($S = 4$), 901.3 ($S = 8$) and 1 664.9 mm (criterion A), because Box3D runs 4 substeps **plus 8 position iterations** per step and our step has none [build/U76/RESULTS.md §2, §6.1]. The remaining cause is therefore the missing position correction, not the time-step size on its own. Third, supplying that correction moves scene #1 from pen/R 7.136 to 1.516 (Box3D 0.706) and the count to 12 of 39 under the old contact cache, and exchanging the velocity solver does not carry the rest: ADMM at $S = 4$ with NGS repairs 8 of 39 with 1 regression and takes scene #1 back to 7.264, while the ablations ($S = 1$: 2 of 39; no NGS: 6 of 39) put the substeps, not the solver, in the load path. All four of those rows ran under the defective contact cache; under the fixed contact path the measured counts are 6 of 39 (PGS) and 7 of 39 (ADMM), 1 regression each [build/U81/RESULTS.md §summary; build/U97/RESULTS.md §8; build/U113/RESULTS.md §“Torn och U23”].

Two sub-results separate out, and this section is frozen at the numbers below.

The 8-part tower is closed. Under the fixed contact cache and four substeps, both velocity solvers give 0 of 8 parts drifting more than 1 mm: PGS at max drift 0.252782308 mm, max penetration 0.0891378235 mm and max residual 7.52658652e-05; ADMM at 4.03233004e-10 mm, 7.43710649e-11 mm and 1.29753361e-11 [build/U113/RESULTS.md §“Torn och U23”]. At $S = 1$ the same tower gives 2 of 8 for PGS (max drift 1.46832219 mm) and 0 of 8 for ADMM (6.52544685e-09 mm), each reproduced over two runs with equal sha256 [build/U113/RESULTS.md §“U97:s tornrecept, $S=1$ ”]. The earlier figures for this result — ADMM with a spectral ρ and an exact in-kernel device Cholesky at 6.525e-06 mm drift against 1 of 8 for ADMM with Ruiz scaling, 4 of 8 for PGS-100 and 4 of 8 with NGS alone — were produced under the old contact cache [build/U97/RESULTS.md §6, §11; build/U81/RESULTS.md §4].

The packing is not closed, and the cause is measured. Run from the initial state with four substeps under the fixed cache, $N = 300$ leaves PGS-400 at a maximum natural residual of 0.0034598158 against a 1e-6 requirement, kinetic energy 0.0328229386 J at $t = 1$ s and 305 of 960 substeps at tolerance, with rest not reached; ADMM diverges at $t = 0.345833333$ s at 1.9152845e+83 J, first residual above 1e12 at macro-step 82 with $\rho = 0.0306066773$. $N = 100$ behaves the same way: PGS 0.0030221998 J at $t = 1$ s and no rest, ADMM divergence at $t = 0.3375$ s [build/U113/RESULTS.md §“Högen från initialtillstånd”, §“Divergens och körgränser”]. The pre-fix measurement stands beside it: 0 of 60 substeps at $N = 100$ and 0 of 20 at $N = 300$ reach the 1e-6 relative tolerance, the residual parks at 1.8e-01 and 2.5e+00 and does not move between 400 and 1 600 sweeps for PGS or for ADMM, the Ruiz rule returns $\rho = 0.1$ at which ADMM diverges (5.9271e+37 and 1.0257e+60), and at $n_c = 9$ 643 the dense Delassus matrix is 6.235 GiB on the host and 145.5 s per substep to upload, so $N = 1$ 000 through the ADMM branch is unmeasured, not green [build/U97/RESULTS.md §summary, §3]. The contact set is not the remaining cause: on more than 2 800 clean contacts the velocity solvers plateau (PGS) or diverge (ADMM) [build/U113/RESULTS.md §summary].

The contact cache was defective and is fixed. With every body held still (largest point displacement 5.7219585e-17 m), a cache hit under the old implementation changed 1 302 of 2 621 body-body normals by more than 1°, 549 by more than 90°, and the gap by up to 23.8745753 mm [build/U113/RESULTS.md §“Felet och rättningen”; build/U111/RESULTS.md §“Cacheuppdatering utan rörelse”]. The cause is lost probe provenance: points probed A→B with normal sign +1 and B→A with sign −1 are stored in A's frame without the probe's origin and sign, the cache update then always queries B's SDF and uses its gradient with sign +1, and it replaces the narrow-phase patch normal with a point gradient; the broad-phase pair's existence does not test whether the contact geometry still holds [build/U113/RESULTS.md §“Felet och rättningen”]. Rewritten to hit only on identical positions, quaternions, shape identities, surface samplings and contact parameters, the same test gives 0 normals off by more than 1° and 0 gap difference, matching the run with the cache disabled [build/U113/RESULTS.md §“Felet och rättningen”].

Every penetration number quoted above for $N = 100$ and $N = 300$ from U44b, U81 and U97, and the U81/U97 rows of the repair ladder, were produced under the old cache. $N = 1\,000$ is a four-substep probe from a checkpoint at macro-step 455, not a run from the initial state and not a rest test [build/U113/RESULTS.md §“Modal $N=1000$ ”]. Re-run through the control pipeline under the fixed cache, PGS repairs 6 of 39 losses with 1 regression among the 39 wins and ADMM repairs 7 of 39 with 1 regression; that lane states that its variants A and B share one implementation in the U23 and tower code path, so it is not a two-variant cache experiment [build/U113/RESULTS.md §“Torn och U23”]. A separate cleanup of the contact set — 0.5 mm gap filter, merging within 1 mm, at most four points per body pair — takes the set from 2 842 to 1 823 contacts, pairs carrying more than four points from 23.3 % to 0 %, and $\kappa^+(G)$ from 22 944 to 5 372, without reaching the residual target [build/U111/RESULTS.md §summary, §“Kontaktmängd och spektrum”].

Two ported recipes were measured; neither closes the packing. A soft-step port of Box3D’s recipe — four substeps, one biased sweep, one relaxation sweep, contact frequency $f = \min(30, 0.125/h)$ Hz, $\zeta = 10$, warm start through the fixed contact matching, run through our exact cone projection and our contact pipeline — repairs 8 of 39 U23 losses with 0 regressions, and leaves the $N = 300$ pile at 0.24039997 J and 7.21229016 mm final penetration at $t = 1$ s with rest not reached. On the 8-part tower it lands at pen/R 0.369196888 against the oracle’s 0.191871709, a relative deviation of 92.4 %, with no rest in our run against the oracle’s 0.645833333 s, and the $N = 1\,000$ lattice deviates by 5.88 relative on a censored window. On that same 8-part tower — the tower this section calls closed for the two velocity solvers — the ported recipe leaves **8 of 8 bodies drifting more than 1 mm, maximum 40.7747524 mm**, maximum penetration 9.10093576 mm, reproduced over two runs [build/U129/RESULTS.md §“Torn, orakel och paritet”]. The normal update is our projected soft NCP with de Saxcé and not an algebraic identity with Box3D’s separate normal and manifold friction solve, the cone is projected exactly rather than as a pyramid, no sleeping is used, and the relaxation pass uses the substep’s frozen Jacobians while Box3D updates anchor separation from position deltas, so the deviation is not isolated to softness [build/U129/RESULTS.md §summary, §“Recept och avgränsning”, §“Högar”, §“U23”, §“Torn, orakel och paritet”]. A barrier port — $b(d) = -(d - \hat{d})^2 \ln(d/\hat{d})$, $\hat{d} = 0.001$ m, $\kappa = 1e5$ N/m, lagged Coulomb-Huber friction with $\varepsilon_v = 1e-5$ m/s, Newton ≤ 10 with block-Jacobi PCG ≤ 200 and contact Jacobians frozen within the step — brings a single cube to rest in 0.0333333333 s and a 1000:1 column in 0.075 s, and on the 8-part tower with 0.5 mm start gaps rests at 0.1833333333 s with 5 of 8 bodies drifting more than 1 mm (max 2.623641491 mm). The touching tower is rejected at the start, where the barrier is undefined at $d = -2.220446049e-16$ m; both piles stop at $t = 0.3166666667$ s on the Newton budget with rest not shown; 5 of 39 U23 loss scenes complete and 34 abort; and on L4 the $N = 1\,000$ pile reaches 0.3041666667 s within the wall budget, censored. CCD, the geometric Hessian and the friction fixed-point iteration are absent, so this is an IPC-type approximation and not the published algorithm, and its Newton stop is $\max |\text{gradient}| < 1e-10$ in energy units, not a physical residual [build/U130/RESULTS.md §summary, §“Implementation och begränsning”, §“Kub, kolumn och torn”, §“Högen och U113”, §“U23, samtliga förlustscener”, §“Modal L4”]. The packing therefore stands unsolved with the cause measured: the velocity solvers plateau or diverge on more than 2 800 clean contacts, and the oracle’s recipe is not reproduced in our frame.

That is the class the router of §2.6 is built for (1-D mass-ratio chains route to PGS + arc-freeze, 423×; packings to block-Ruiz + ADMM); the re-run of all 39 losses through the router is reported in §4.9 and is worth one scene over the PGS-100 control.

All channels are bit-identical over repeated runs: the U23 GPU channel 98b7369232d9012838b18a3d1cffe8020b899987c5944896882c0aac6832e233 and CPU channel 1e8de67f0d577cc6aa659193697339aa916faa799a30b40a4e7f85812cc816c1 [build/U23/RESULTS.md §summary]; 10 of 10 substepping configurations over two processes and over the local RTX 5070

against a Modal L4 (ap-nVa54QeZa1M6y1FABFFKLZ, 657.7 s = 10.96 GPU-minutes) [build/U76/RESULTS.md §5]; and both NGS runs at every N [build/U81/RESULTS.md §3]. The NGS projection changes nothing on non-penetrating scenes: the pushed cube’s and the 3-box column’s impulses are byte-identical with and without it [build/U81/RESULTS.md §summary].

4.8 External anchors

Correctness is pinned to closed forms and to two independent engines, not only to internal consistency.

- $\Sigma \lambda_n = M g \Delta t$ on the ground interface: relative error $2.8e-13 \dots 3.8e-11$ over six scenes [build/RESULTS.md §A3].
- Pushed-cube a_x against $(F - \mu mg)/m$: $1.4e-15$ [HUNT.md §Mätt/CPU-referens]; spectral-p run $1.95e-13$ with $\Sigma \lambda_n$ at $1.95e-11$ [HUNT.md §Mätt/H15].
- Complementarity at the ADMM solution, $\text{tol } 1e-13$: $\varepsilon_p = 0$ exactly (the cone projection is exact and z is the projected iterate), $\varepsilon_d \leq 4.9e-13$ over six scenes [build/RESULTS.md §A3].
- Box3D (Erin Catto, MIT, -ffp-contract=off) as an external CPU oracle: bit-identical over 1/4/8 threads on 5 scenes — a *stronger* determinism claim than ours, which holds for the same device and launch configuration [HUNT.md §Mätt/Box3D-oraklet]. On the pushed cube our trajectory deviates by 2.0 mm max, path length 0.23 % [HUNT.md §Mätt/Box3D-oraklet].
- MuJoCo 3.9.0 as a second oracle: its finite-difference derivatives differ from our adjoint by 1.25 (dv⁺/dv) and 2.53 (dv⁺/dF) in relative Frobenius error. Its FD is *clean* — spread $1.7e-9$ over three ε — so the difference is **model, not noise**: a convex smoothing against an exact projector. MuJoCo also carries position sensitivity 10.8 in the velocity rows, where our adjoint at fixed J has zero there by construction [build/U6/RESULTS.md §2, §summary].

4.9 Two [MISSING] items closed

The router does not repair the 39 losses, and the batch was not contention-limited. All 448 U23 scenes are on disk with their parameters and anchors, so all 39 losses — not only the top ten — were re-run with the per-step NCP solve going through the U18/U33 CPU router instead of the 100-sweep PGS of the gpu_exact channel, everything else (contacts, integrator, anchor evaluation) unchanged: the router repairs **6 of 39** by U23’s own loss rule (score = slip_err/0.01 + pen/R + 10·tip_wrong, lost if score > 1.5 × the winner’s), 20/39 on the penetration anchor alone and 5/39 on both anchors, against 5/39 for a plain PGS-100 control on the same CPU world — so re-routing is worth one scene [build/P2close/RESULTS.md §A.3]. The branch that fires is the one §4.7 predicted (PGS + arc-freeze on 936 of 1 170 steps, exact-bilateral on 213, an ADMM branch on 5, so the GPU arm never carries this family), but it is not the lever: 481 of 1 170 steps stop at max_sweeps = 300 short of $1e-8$, and raising that cap tenfold on the five worst scenes cuts the unconverged steps (14→8, 9→2, 10→4 of 30) while moving pen/R by less than 0.4 — scene #1 lands at 5.026 / 5.881 / 6.500 / 7.136 for CPU-PGS-100 / router-3000 / router-300 / GPU-PGS-100 against Box3D’s 0.706, a spread that the single-substep discretisation, not the solver, admits, which is the same substepping cause §5.4 already isolates; the §4.7 attribution “Delassus ill-conditioning in the PGS path” is therefore **not supported**. Both passes are bit-identical on every physics field (sha256 c6419081...; the sweep-equivalent cost field is excluded and differs in 2 of 39 scenes because get_chol_ratio falls back to a wall-clock measurement) [build/P2close/RESULTS.md §A.5]. On a dedicated L4 (Modal ap-wxPuQkFLeukBtXUddCbHfh, 81.9 container-seconds = 1.37 GPU-minutes) the ANYmal C stick batch gives **16 704 env-steps/s** (245.22 ms) and **79 507 adjoint directions/s** (103.04 ms) at B = 4 096, 133 launches per ADMM iteration and 0.9971 converged — **0.95×** and **0.959×** the contended local numbers of §4.5, so contention was never the limit; the local RTX 5070 is simply ~5–8 % faster than an L4, matching U20’s 0.92× for A1. Talos (nv 38, n_c 2) reaches **4 920 env-steps/s** and 72 010 directions/s (1 direction/env, all-stick) with a converged fraction of 1.0000 at every B, and it **stops**

scaling between B = 1 024 and B = 4 096 ($\times 4$ the envs for $\times 4.71$ the time, against $\times 2.00$ for ANYmal C) at an identical launch count, i.e. the 38×38 dense M^{-1} matvec makes it compute-bound where ANYmal C is still launch-bound. On both robots 8/8 envs are bit-identical alone versus in the batch over the full sha256 of $(\lambda, v^+, d\lambda/d\tau, dv^+/d\tau, d\lambda/d\mu)$, and the ANYmal C batch hash on the L4 equals the local RTX 5070 hash 3dd2518f... [build/P2close/RESULTS.md §B.1–B.5].

4.10 System identification through the adjoint

The adjoint of §2.3 is used as the Jacobian of a Gauss-Newton estimator. A Talos arm pushes a box 0.20 m in +x over 1.00 s; the box’s mass, its yaw inertia I_{zz} and the box-table friction μ are estimated from the joint torques alone, 512 environments in a batch.

The forward model is full Talos dynamics, not a stance export: $M(q)$, $M^{-1}(q)$, $nle(q, v)$ and $J_hand(q)$ are recomputed with Pinocchio at the integrated state for every env and each of the 100 steps, and the NCP is solved by the batched kernels of §2.7. The natural NCP residual is at most $4.213074e-12$ over 512×100 steps, p50 $3.096e-13$ [build/U98/RESULTS.md §1, §5]. The estimator solves the same NCP with the same kernels on a reduced 24-DOF scene whose (G, b) equal the full 53-DOF scene’s, so estimator and generator are one model and not two: at the true parameters the Gauss-Newton cost is $2.422259e-06$ over 512×100 steps $\times 32$ residuals at $\sigma = 0$, and $2.347430e-06$ over the 50-step window [build/U98/RESULTS.md §6].

Two modelling choices are stated rather than buried. Four coplanar corner contacts make the Delassus operator rank-deficient — 15 rows, 9 independent, $\text{cond}(G) = 3.25e17$, and two solvers return $|\lambda_PGS - \lambda_ADMM| = 1.5e-1$ on different active sets at $\varepsilon = 0$ — so a linear contact compliance $\varepsilon = 0.02$ is used, realised as 15 extra generalized coordinates with $M^{-1} = \varepsilon I$ and $J = I$ on those columns, which gives exactly $G_real + \varepsilon I$ without touching any solver or the scene contract [build/U98/RESULTS.md §2, §11]. And an errors-in-variables form was measured and dropped: feeding the measured torque into the hand’s v_free multiplies σ by $dt \cdot |J_h M(q)^{-1}| \approx 5$ (m/s)/(N·m) on Talos’s lightest arm joints, and gave m p95 = $1.6e-1$ at $\sigma_\tau = 1$ N·m with NCP residuals up to $9.9e4$ inside the Gauss-Newton loop; every number below comes from the maximum-likelihood form that conditions on the measured joint velocities [build/U98/RESULTS.md §11].

The adjoint is checked against central differences before it is used: relative p95 $1.774e-07$ for m and $2.044e-07$ for μ at $h = 1e-5$, and for μ the derivative is exactly 0 in the 94 of 256 samples where no floor contact slips, with $|FD| \leq 4.479e-08$ there [build/U98/RESULTS.md §6].

The estimate lands on the Cramér-Rao bound. $F = J^T J / \sigma^2$ is formed from the same contact adjoint at the true parameters; the expected median of $|\text{error}|$ for an efficient estimator is $0.6745 \cdot \text{CRB} \cdot \sigma$.

window	parameter	CRB rel per N·m (p50)	σ_τ [N·m]	$0.6745 \cdot \text{CRB} \cdot \sigma$	measured p50
20	m	0.1462	1.0	$9.8634e-02$	$9.3553e-02$
50	m	0.0573	1.0	3.8621e-02	3.8590e-02
100	m	0.0351	1.0	$2.3649e-02$	$2.2260e-02$
50	μ	0.0594	1.0	$4.0072e-02$	$4.1072e-02$
50	m	0.0573	0.1	$3.8621e-03$	$4.0476e-03$
50	μ	0.0594	0.1	$4.0072e-03$	$4.3778e-03$

The agreement holds in all nine cells of parameter $\times$ window [build/U98/RESULTS.md §7]. The error is monotone in the noise in 9 of 9 parameter/window combinations, and the $\sigma = 0$ floor is $1.2e-06 \dots 2.5e-04$ relative for m and μ , so no part of the error is a model gap between generator and estimator [build/U98/RESULTS.md §8.2].

Three acceptance rows fail, and the measurement says they fail on information, not on the estimator. p95 < 0.05 at $\sigma_\tau = 1$ N·m over 50 steps is not met — $1.5173e-01$ for m and $1.6086e-$

01 for μ — while the CRB p50 there is 5.726e-02 per N·m, so an unbiased estimator's p95 sits near 1.4e-01; the same criterion is met at $\sigma_\tau = 0.1$ N·m (1.5272e-02 and 1.6340e-02) and at 100 steps with $\sigma_\tau = 1$ N·m (8.9702e-02 and 8.3003e-02) [build/U98/RESULTS.md §8.1, §10]. Starting at the true parameters instead of the neutral start moves m p95 from 1.5173e-01 to 1.4101e-01, so the optimizer is not stopping in local minima [build/U98/RESULTS.md §8.3]. And the Fisher gate does not separate the misses: $-\log_{10} \lambda_{\min}(F)$ gives test AUC 0.6108 at $\sigma = 1$ and 50 steps (0.7828 at 100 steps), a CRB score from the same F gives 0.7157, and an oracle CRB evaluated at the **true** parameters gives 0.7141 — the flag sits at its own information ceiling and 0.9 is not reachable here [build/U98/RESULTS.md §8.4].

Two baselines at the same σ and window: a quasi-static $F = m \cdot a$ fit from the hand force gives m p50 3.4138e-01 against 3.8590e-02, and a recursive μ estimator fed the quasi-static mass gives 2.6829e-01 against 4.1072e-02. Fed the **true** mass the same recursive estimator gives 1.3501e-02, below the joint estimate, because Gauss-Newton estimates three parameters at once and the m- μ correlation lifts the μ error [build/U98/RESULTS.md §8.5].

Cost and determinism on the same scene: at B = 4096 it runs 15 008.0 env-steps/s and 41 908.9 adjoint directions/s on the shared local GPU, two batch runs give sha256 33661948..., env-in-batch equals env-alone in 8 of 8 environments, and the whole generator gives 75b64d26... in two separate processes [build/U98/RESULTS.md §9]. The identified parameters are simulated ground truth, not measured material properties (§5.5).

Estimation inside the step loop. A per-env μ estimator written as a Warp kernel with int64 reduction, called after every forward step, costs -0.05% throughput at B = 4096 on a dedicated L4 (forward -0.04% , gradient -0.14%), and its kernels sum to 18.783 μ s, 0.00402 % of one forward step; the gradient graph itself contains no estimator update. The batch hash is the same on the L4 (sm_89) and on the local RTX 5070 (sm_120), 555bf88fa00d6a19d14c6e517822e7ccb886a32fb9852e5831c2ddefc19de9e8, and env-in-batch equals env-alone in 8 of 8 over $(\lambda, v^+, d\lambda/d\tau, dv^+/d\tau, d\lambda/d\mu, \hat{\mu}, \sigma, \mu_{\text{eff}})$ [build/U80/RESULTS.md §summary; build/A64/RESULTS.md]. The fall-rate comparison in that lane (3.83 % at k = 1 against 19.14 % with known μ at step 50) is **not** an isolated estimator win: margin, probing and solver convergence enter the comparison, known μ gives 784 falls without margin, 180 with k = 1's σ series and 0 with its own σ , and the explanation that known μ fails below $\mu = 0.35$ is refuted — the falls sit at $\mu > 0.6$ [build/A64/RESULTS.md].

Two earlier lanes are simplified pre-studies of this measurement, and are quoted as such. One estimates μ , a trunk mass scale and restitution jointly on ANYmal C (p95 at a 20-step window 0.0283 / 0.0407 / 0.0053, rising to 0.0971 for μ at a 50-step window, against 0.1572-0.4359 for a recursive μ -only baseline) [build/U84/RESULTS.md §summary; build/A67/RESULTS.md §2]; the other estimates all 16 friction tiles of a 2 m $\times$ 2 m terrain at once (16 of 16 below a 0.05 gate, median 0.0011, max 0.0074 at 4096 envs, with the Gauss-Newton system accumulated in int64 fixed point across threads and envs) [build/U92/RESULTS.md §summary]. Both run a contact-step model whose M and J are held fixed from a stance export instead of being recomputed at the integrated state, so neither is full robot dynamics [build/A67/RESULTS.md; build/A69/RESULTS.md]. The terrain lane's Fisher = 0 and $\mu = 0.5$ on the two unvisited tiles are written in after the estimation rather than measured [build/A69/RESULTS.md]. The ANYmal C lane's Fisher threshold finds 9 of the 11 environments that miss the μ gate, misses environments 520 and 570, and flags 93 environments that pass it [build/A67/RESULTS.md]. Its restitution row identifies the normal factor of that contact-step model; physical restitution in gait is not validated [build/A67/RESULTS.md].

Excitation design moves the mass bound, not the friction bound. The push trajectory was parameterised by five numbers — push amplitude A, min-jerk time T, side amplitude A_y, side reversals n_y per 0.50 s window and the contact lever y_{off} — and 40 scrambled-Sobol candidates

were screened with the estimator code above, imported rather than copied, at the true parameters. D-optimal (max logdet F_{rel}) and E-optimal (max $\lambda_{\text{min}} F_{\text{rel}}$, taken with the graph engine's identifiability_oed) select the same candidate among those whose generator solves the NCP to the accuracy of the standard trajectory, both scores taken on the (m, μ) block of the Fisher in relative parameter units [build/U116/RESULTS.md §summary, §4, §5]. Against the standard trajectory at $B = 512$, $\sigma_{\tau} = 1 \text{ N}\cdot\text{m}$ and 50 steps, that candidate lowers the CRB p50 for m from $5.725943\text{e-}02$ to $1.713034\text{e-}02$, a factor 3.343, and for μ from $5.941064\text{e-}02$ to $3.434491\text{e-}02$, a factor 1.730; the measured p95 error for m goes from $1.517289\text{e-}01$ to $3.818469\text{e-}02$, below 0.05, and for μ from $1.608568\text{e-}01$ to $8.413782\text{e-}02$, above it. The standard column reproduces the published numbers digit for digit before anything new is measured [build/U116/RESULTS.md §7]. The μ bound belongs to the family and not to the selection: none of the 40 candidates reaches a CRB p95 for μ below 0.05, the lowest being 0.0664, while three reach it for m [build/U116/RESULTS.md §summary, §6]. The mechanism is measured. Information about m comes from the inertial term and grows with acceleration — that candidate runs max $|a_{\text{ref}}| = 169.2 \text{ rad/s}^2$ against the standard's 5.0 — while information about μ comes from the friction force $\mu \cdot m \cdot g$, whose normal force is the box's weight regardless of acceleration, so the only route to more μ information is more slipping steps, and it loses hand contact in 72.13 % of steps against the standard's 0 % [build/U116/RESULTS.md §7]. It also costs $5.284\times$ the URDF torque limit against the standard's $1.154\times$, box yaw 1.2171 rad against 0.1093, and a Gauss-Newton cost at the true parameters of $1.503296\text{e+}01$ against $2.347430\text{e-}06$ (both over the 50-step window) with an estimator NCP residual of $1.481804\text{e-}02$ against $1.282599\text{e-}06$, so on that trajectory the estimator's model is no longer the generator's model. Within the standard's own torque ratio 1.0567 the best CRB ratios in the family are 1.253 for m and 1.352 for μ ; among the 8 candidates that never lose contact they are 1.554 and 1.809 [build/U116/RESULTS.md §7, §8].

5 Limitations

5.1 The x-step PCG is the remaining GPU-vs-CPU gap in the ρ rules

The ρ rules of §2.5 were derived and validated on a CPU router with an exact Cholesky x-step. On the GPU the x-step is 16–64 PCG iterations, and that is where the two diverge [HUNT.md §Mätt/U9]. Measured: on the sliding column ($n_c = 3$) an exact device Cholesky takes 76 iterations in 42.88 ms, **3.3× faster** than CG = 16 (141.3 ms) and $44.2\times$ faster than CG = 256 (1 897.3 ms) [build/U33/RESULTS.md §summary]. On packings, CG = 16 converges in 436 and 628 outer iterations against the CPU Ruiz path's 565 and 577 ($1.09\times$), while CG = 64/256 adds $1.4\text{--}2.7\times$ compute for no reduction in outer iterations [build/U33/RESULTS.md §summary]. The earlier statement that no $\eta > 0$ packing converges through the PCG x-step is **refuted** by a later re-measurement: with PCG-16 and the Ruiz ρ , pack50 converges in 390 sweeps to $9.1911374535\text{e-}09$, dem_step in 480 sweeps to $9.79868054014\text{e-}09$ and pack800 in 360 sweeps to $9.93350818545\text{e-}09$, all under the $1\text{e-}8$ gate [build/A60/RESULTS.md §1, §5; HUNT.md §Mätt/U9]. The worst router ratio (col_slide $49.5\times$) stands [build/U33/RESULTS.md §summary].

One defect in the GPU Ruiz branch is open: the generated clamp uses the interval $[1, 1]$ and moves the norm of G by up to 25.59 % ($25.5525 / 25.5806 / 25.5907$ % on pack50 / dem_step / pack800), so the runtime η does not follow the stated spectral norm in the compiled path. An isolated copy with the bounds set to ± 1 takes the maximum relative norm error on pack50 from 0.255525 to $6.95173\text{e-}09$; no result in this paper uses that intervention [build/A60/RESULTS.md §10, §U55]. A re-measurement of the bil_A path on pack800 is UNVERIFIED after an 85.131 s timeout [build/A60/RESULTS.md §U55]. The router's GPU arm therefore inherits an accuracy knob the CPU arm does not have, and one open norm defect.

5.2 The A1 regime: small n_c , articulated, floating base

Small articulated systems are a fourth regime, and ADMM is the wrong method there: $n_c \leq 4$ floating-base articulated scenes converge in 27–53 PGS sweeps where the ADMM ladder spends 209–485 [build/U18/RESULTS.md §summary]. Before the branch was added, three A1 scenes were the worst-routed in the benchmark ($9.15\times$, $7.76\times$, $6.04\times$) [build/U33/RESULTS.md §summary]. The branch that fixes them was selected on the same 21 scenes, and the summary does not report its leave-one-out error [HUNT.md §Mätt/U33].

Per-step latency at this size is not competitive and the reason is structural: at $n_c = 4$ we are $\sim 1500\times$ slower per step than Pinocchio (16–41 ms against 0.013–0.016 ms), and the measured cost is **launch overhead**, not arithmetic — one uncaptured matvec (7 kernels) takes 108 μ s, i.e. $\sim 15 \mu$ s per launch, and one ADMM iteration (~ 130 launches) takes ~ 1 ms [build/h24/RESULTS.md §5]. The GPU claim is the batch, not the single scene — and even there, Pinocchio single-core does 4096 A1 scenes in 174 ms against our 288.62 ms, without a cone and without $d\lambda/d\mu$ [build/h25/RESULTS.md §Verdict row 7].

Six of 4096 all-stick envs diverge (to $1e+55$) at $\rho = \lambda_{\min}^+$ although the certificate proves all-stick; all six are in the bottom 0.3 % of minimum normal impulse, so the rule needs a near-open third branch [build/h25/RESULTS.md §Verdict row 6; HUNT.md §Mätt/H25]. That third branch’s threshold τ is **unidentifiable on the 13-scene router set** and needs the 4096-env population to fit [HUNT.md §Mätt/H23b].

5.3 Packings need η , and no spectral rule covers them

On granular packings nothing converges at $\eta = 0$ — the residual parks at $5e-7 \dots 4e-5$ [HUNT.md §Mätt/H18]. $\rho = \lambda_{\min}^+$, which wins on chains, towers and lattices, is refuted on packings, where $\rho \approx 1.1-2\sqrt{(\lambda_{\min} \lambda_{\max})}$ is required, and **no covariate found separates the two families**; $\text{cond}(G)$ does not [HUNT.md §Mätt/H1b, §Mätt/H18]. The measured fix is a different route rather than a different constant: block-Ruiz equilibration plus a constant $\rho = 0.1$ gives 627.966 sweep-equivalents on pack800, $0.678\times$ the 926 iterations at U135’s 25-point sweep oracle ρ , *without any spectrum, and does **not** transfer to chains* [HUNT.md §Mätt/H23b]. *A physically derived η from Hertz contact [R18b] was tried and makes packings harder** (pack50 533 $\rightarrow$ 1044; pack200/pack800 do not converge) [HUNT.md §Mätt/H33].

5.4 Box3D and stiff MuJoCo hold the 1000:1 column better at small N

engine	final z (m)	max penetration	shoot-out	source
MuJoCo stiff	[0.09993, 0.29985, 0.49978]	6.39e-4 m	no	build/U6/RESULTS.md §1(iii)
Box3D, 16 substeps + 240 Hz	[0.09892, 0.29029, 0.48597]	1.15e-2 m	no	build/U6/RESULTS.md §1(iii)
Box3D default	[0.05290, 0.08515, 0.14912]	1.90e-1 m	no (sinks, stops)	build/U6/RESULTS.md §1(iii)
ours, exact, 800 it	[0.09885, 0.29698, 0.49533]	4.45e-3 m	no	build/U6/RESULTS.md §1(iii)
ours, exact, 200 it	[0.10757, 0.12307, 0.10000]	2.17e-1 m	yes (xy -1.96 m, KE 3.88 J)	build/U6/RESULTS.md §1(iii)

MuJoCo stiff is the tightest of all at 0.64 mm and 0.03 ms/step; Box3D never shoots out at any budget, while we do at 200 iterations [build/U6/RESULTS.md §1(iii); HUNT.md §Mätt/Box3D-oraklet]. Our 800-iteration answer is $2.6\times$ tighter than tuned Box3D and $2\,700\times$ more expensive [HUNT.md §Mätt/Box3D-oraklet]. Scaling on the same machine: $N = 1\,000$ Box3D 0.45 ms against our 2.07 ms, $N = 10\,000$ 3.30 against 3.08, $N = 40\,000$ 14.4 against 7.3, crossover $\approx N = 15\,000$ [HUNT.md §Mätt/Box3D-oraklet]. That crossover is a **hardware-pairing number, not a**

property: in the same container against 8 Box3D workers it moves to $N^* \approx 3\,800$ ($\approx 1\,240$ against one worker), and it must always be quoted with the pair [HUNT.md §Mätt/Modal].

Related model limits, stated: contact generation has no edge-edge points (vertex-plane plus SAT box-box with reference-face clipping, ≤ 4 points per face pair), which is adequate for stacked boxes and not for general tumbling boxes [build/RESULTS.md §A1]. On hyperstatic scenes λ is genuinely non-unique — $\text{rank}(G) = n_v$ on every box scene — and only u and the generalized force $J^T \lambda$ are well-defined; the sensitivity routine flags `singular=True` and falls back to the pseudo-inverse rather than returning a number [build/RESULTS.md §A2, §A5].

5.5 No real force sensor

Every anchor in this paper is analytic ($\Sigma \lambda_n = Mg\Delta t$, $v_0^2/2\mu g$, $a_x = (F - \mu mg)/m$) or another simulator (Box3D, MuJoCo, Pinocchio). **No measurement against a hardware force/torque sensor or a physical robot exists in this campaign.** The friction coefficients the stance controller estimates are simulated ground truth, not measured material properties [MISSING — see §6.3]. The slip-probe controller has been run quasi-statically in simulation only; its torque budgets ($\Sigma |\tau| = 38\text{ N}\cdot\text{m}$ on A1, $358\text{ N}\cdot\text{m}$ on ANYmal C against an $80\text{ N}\cdot\text{m}$ per-joint limit) are feasibility numbers, not hardware results [build/h30/RESULTS.md §0; build/U19/RESULTS.md §summary].

5.6 Determinism is device-scoped, and one host gap stays open

Our bit-identity holds for the same device and the same launch configuration; Box3D’s holds across thread counts on a CPU, which is the stronger claim [HUNT.md §Mätt/Box3D-oraklet]. Cross-machine identity requires `fuse_fp=False`, `fast_math=False` **and** a matched host BLAS microkernel; `OPENBLAS_CORETYPE=Haswell` closes the `np.dot/np.linalg.inv` channels, but `eigvalsh` (`dsyevd`) is not fixable cheaply and leaves p^* differing by 111 ulp (cube) and 33 ulp (tower) [HUNT.md §Mätt/fuse_fp-fix; build/digests/RESULTS.md §0b]. Two confounders remain formally open: CPU dispatch against Python version, and architecture against `nvrtc` version against driver [build/digests/RESULTS.md §0 / §7]. Two of 43 gates fail under the contraction-free default and are left failing [build/RESULTS.md §B.i.1].

5.7 Comparisons this work does not make

Each row below names a measurement a reviewer can ask for and that no lane in this campaign ran. None of them is scheduled here; they are stated as the boundary of what the numbers above cover.

comparison	what exists instead	why it is a gap
Gradients against MJX or Brax autodiff. No run of either exists in this campaign.	The adjoint is compared against MuJoCo’s finite differences (<code>mjd_transitionFD</code> , relative Frobenius 1.25 on dv^+/dv and 2.53 on dv^+/dF , §4.8) and against Pinocchio’s analytical bilateral derivatives ($1.2e-13 \dots 1.3e-12$ on $dv^+/d\tau$, §4.4) [build/U6/RESULTS.md §2; build/h24/RESULTS.md §4].	MJX is a JAX implementation differentiated by reverse-mode autodiff through the solver iterations [02_sota_ncp_mpm.md §A6], and that is the GPU-batched gradient the diff-sim literature compares against. Brax appears in neither the source map nor this paper.
Gradients against Dojo on one shared scene. No run of Dojo exists in this campaign.	Dojo is described in §1.1 from the source map only [02_sota_ncp_mpm.md §A4].	Dojo’s claim is analytical gradients through a hard NCP, which is the same claim as §2.3.
Reverse-mode autodiff of the unrolled ADMM iterations (wp.Tape) on the same scene. Not run.	Central finite differences $3.94e-8$, an independent CPU implicit-function reference $1.7e-14$, and Pinocchio (§4.4) [build/h1/RESULTS.md §Verdict].	The solver is written in Warp [R12], which carries the tape, so this is the cheapest direct check that the adjoint is the forward operator transposed, and it would put a number on §2.3’s statement that no second solver exists.

comparison	what exists instead	why it is a gap
The determinism price against another deterministic reduction. Not run.	int64 fixed point against float32 atomics: 2.40–2.46× before fusion, 1.103–1.161× after [build/h6/RESULTS.md §Verdict; build/U3/RESULTS.md §summary].	§1.3 cites the literature’s deterministic alternatives — segmented sort-based reduction, per-body fixed-order reduction, compensated summation — as costing throughput [02_sota_ncp_mpm.md §“Known hard problems” item 3]. None was measured, so 1.10–1.16× has no competitor.
N = 1 000 packing from the initial state. Not run.	N = 100 and N = 300 were re-run from the initial state under the fixed contact path (§4.7); N = 1 000 is a four-substep probe from a checkpoint at macro-step 455 [build/U113/RESULTS.md §“Modal N=1000”].	The checkpoint carries the old simulation’s body history, so the probe measures neither time to rest nor a run from the initial state.
Batched throughput on a cone-binding batch. Not reported.	17 047 env-steps/s and 120 393 adjoint directions/s are measured on an all-stick batch, 0.9985 of it below 1e-8 in 200 iterations [build/U20/RESULTS.md §summary].	All-stick is the regime the certificate of §2.6 routes <i>past</i> ADMM. On the cone-binding batch 77.7 % reach 1e-8 in 2 000 iterations, q90 = 1.33e-7 and max = 1.09e-4, and no throughput is reported for it [build/h25/RESULTS.md §Verdict row 4, §4].
Uncontended throughput on ANYmal C and Talos. Partly run.	ANYmal C 17 561 env-steps/s and Talos 4 920 env-steps/s are contended local numbers; only the A1 batch and the ANYmal C stick batch were re-run on a dedicated L4 [build/U19/RESULTS.md §summary; build/P2close/RESULTS.md §B.1–B.5].	Local millisecond numbers in this campaign carry a measured spread of up to 2.7× [build/oracle/RESULTS.md §0].
A like-for-like batched comparison against Kamino [R1] and MJX [R6]. Not possible from the sources.	Neither publishes steps/s in the source material [HUNT.md §Mätt/H25].	Stated so the 17 047 env-steps/s is not read as a comparative number.
Hardware. No force/torque sensor and no physical robot anywhere in the campaign (§5.5).	Analytic anchors and two external simulators.	The friction coefficients the stance controller estimates are simulated ground truth.

6 Reproducibility

6.1 Digests

Every bit-identity claim carries a full sha256 of the full array, hashed as `sha256(np.ascontiguousarray(a).tobytes())`, measured twice locally [build/digests/RESULTS.md §0; build/U20/RESULTS.md §2]. Host-built inputs and device outputs are hashed **separately** — 45 host inputs (Ja, Jb, bias, M^{-1} , v_{free} , folded over every step; for the ADMM path also G, G+E, eigvalsh and ρ^*) against 22 device outputs — so the certificate names the side that diverges [build/RESULTS.md §B.i.3].

Selected anchors:

object	sha256	source
pushed cube, 400 it, 250 steps, trajectory	0afe055c3d5514c4487f8276c479a4462564bc208f897e71114a55f4d15b0a01	build/RESULTS.md §B.i.3
K8 tower, 200 it, 600 steps, trajectory	ce0eeafd9f293ff953c8a2fb0610e309242bf84b32e79cf8da3e07fe2baf6536	build/RESULTS.md §B.i.3
lattice N = 10 000, 10 + 30 steps, state	3cddd8b07143ecf44b0fd56d4ccc011561d0ec272251f7f537c933dc981561b3 (identical on 5070 and L4 under fuse_fp=False)	build/RESULTS.md §B.i.3

object	sha256	source
4096-env batch, stick (λ , v^+ , $d\lambda/d\tau$, $dv^+/d\tau$, $d\lambda/d\mu$)	4d5d4747983a3de20aed4d49b12f453ced005275bce8edd94e956d4aa9c3c7d3	build/U20/RESULTS.md §2.1
4096-env batch, slip	24a370fedefcc7a7b2a3dddb7ebe1bc2432a0259bc2c2265304cb4db1bfd9bf6	build/U20/RESULTS.md §2.1
A1 λ / gradient / q-gradient (Pinocchio parity)	f39aed82de5fb535... / e5d63a5c51ec0603... / 33b13dffc77203f4...	build/h24/RESULTS.md §5
ANYmal C λ / gradient	06c28cf6b850171a... / ca966ae76297c40e...	build/U19/RESULTS.md §1
Talos λ / gradient	0157f45f9d215434... / 018e03440ffb0764...	build/U19/RESULTS.md §1
A1 stance controller, rolling digest over 16 solver calls (local)	11ffd878a057560dc61e5e110a5af129f62e3bed343164d9fa71b459d677d097	build/h30/RESULTS.md §0
the same rolling digest on L4 = A10G = H100	9d4ce2cc2904e89912627c7dd5e0b06a5350b7a1295dd5ac5c13f195a8e60938	build/U20/RESULTS.md §2.2

The controller's local/cloud difference is attributed, not waved away: `h30_ctrl.py` runs an adaptive Fisher/belief update in NumPy on the CPU (eigvalsh inside $\rho = 0.20 \cdot \lambda_{\max}$, and a min-norm null(J) projection), which is precisely the OpenBLAS channel of §3.3; all three cloud architectures produce the same state on the GPU [build/U20/RESULTS.md §2.2].

The 10 published MPM regression hashes are identical across `sm_86`, `sm_89` and `sm_90` and differ from `sm_120`, which is the expected consequence of `nvrtc` 12.8 against 12.9 and is recorded as such [build/U20/RESULTS.md §2.5].

6.2 Certificates

A per-run certificate stamps `fuse_fp`, `fast_math`, `Warp`, `Python`, `NumPy`, `nvcc`, `driver` and `sm` version, plus the **host** SIMD family and `OPENBLAS_CORETYPE`, and hashes host inputs and device outputs separately [build/cert/RESULTS.md §summary]. Its gate rows are: bit-repeat within one process and across two processes; float32–float64 parity (measured gap $1.92\text{e-}8$ against a $1\text{e-}6$ gate); the `-ONE` float64 constant (a Warp 1.13 defect where `-ONE` compiled to `+1.0` and shortened ρ by 4 % in the float64 build); and fail-closed NaN injection (`residual(nan) = nan`) [build/cert/RESULTS.md §summary; HUNT.md §Mätt/U4, §Mätt/B3]. A stamped instance: `fuse_fp=False fast_math=False, Warp 1.13.0, Python 3.10.12, NumPy 2.2.6, nvcc 12.8.93, driver 580.178.04, sm_120, host Haswell/AVX2 on an i5-13600K` [build/cert/RESULTS.md §summary].

Applying the same gate to all new kernels gives 6/6 bit-repeat, with three rows recorded red and left red: NaN injection on the MPM int64 P2G (which absorbs NaN to $-8.9\text{e}9$ J instead of propagating it — a formulation-class failure), and f32/f64 parity on the ADMM solver ($2.6\text{e-}2$) and MPM ($7.4\text{e-}4$), both tolerance-class [HUNT.md §Mätt/U36].

6.3 Environments and cloud jobs

Local: RTX 5070 12 GB, driver 580.178.04, `sm_120`, CUDA 12.9, Warp 1.13.0, NumPy 2.2.6, Python 3.10.12; host Intel i5-13600K (AVX2). Every local millisecond in the archived lanes carries a shared-GPU caveat with a measured spread of up to $2.7\times$; the uncontended numbers of §4.3 and §4.5 supersede them [build/oracle/RESULTS.md §0; build/U20/RESULTS.md §summary].

Cloud: Modal client 1.5.0, Ubuntu 24.04, Python 3.12.1, Warp 1.13.0, NumPy 2.2.6, driver 580.95.05 on L4, A10G and H100 [build/U20/RESULTS.md §0].

job	device	Modal app id	stamp	container s	source
smoke (recipe check)	L4	ap-iueeUTkookE9X0xEyKaDb1	20260919T135936Z	44.8	build/modal/RESULTS.md §0
timings, first pass	L4	ap-GITQnxc9jcXEb0HESJhoJx	20260919T140621Z	346.5	build/modal/RESULTS.md §0

job	device	Modal app id	stamp	container s	source
identity	A10G	ap-DQ8bwoP8YnU7wi88RJGcdW	20260919T140626Z	78.1	build/modal/RESU LTS.md §0
identity	L4	ap-zv6mm8VX1z03P4waLwZRYW	20260919T142747Z	91.5	build/modal/RESU LTS.md §0
identity	H100	ap-J0YDRQQ7x6B3NxHoCsGzfk	20260919T142752Z	73.6	build/modal/RESU LTS.md §0
divergence probe, baseline + both codegen toggles	L4	ap-xdW3viDcYLRzzJtWUs9lW2	20260919T145902Z	239.7	build/digests/RES ULTS.md §1
host/device attribution probe	L4	ap-jwWmC0NoEaFI3Lb5v2Foi7	20260919T151236Z	53.9	build/digests/RES ULTS.md §1
ADMM solver under the codegen toggles	L4	ap-bK9UUobTkijz0bbseHyRPC	20260919T151438Z	45.8	build/digests/RES ULTS.md §1
OPENBLAS_CORETYPE confirmation	L4	ap-uubEmWJkSaB7I2nMmC1m7N	—	107.8	build/digests/RES ULTS.md §8
paper timings, part 1 (ncp_gpu, ADMM, batch, controller, patch, joint NCP)	L4	ap-2TSkvydBvCiyDp5a0UE4mw	20260919T215104Z	342.6	build/U20/RESUL TS.md §0
paper timings, part 2 (shape derivatives, MPM, placement)	L4	ap-LaYz8viiWkP0n60II9NVCo	20260919T220048Z	204.3	build/U20/RESUL TS.md §0
cross-device identity, 5 suites	L4	ap-8s59kZejpgFuif4rgwALbN	20260919T220807Z	194.0	build/U20/RESUL TS.md §0
cross-device identity, 5 suites	A10G	ap-6Cmh1Fy9QhJkvSS6r2TL5D	20260919T221435Z	132.0	build/U20/RESUL TS.md §0
cross-device identity, 5 suites	H100	ap-zjX0RSmd9FL3ZnmYRXTs9B	20260919T221955Z	135.1	build/U20/RESUL TS.md §0

Billed: 16.8 GPU-minutes for the paper-number jobs [build/U20/RESULTS.md §0], 13.4 for the first identity pass [build/modal/RESULTS.md §0], 5.7 for the divergence probes [build/digests/RESULTS.md §1], 1.8 for the OPENBLAS_CORETYPE confirmation [build/digests/RESULTS.md §8], 0.24 for the fused-scale A/B [build/U3/RESULTS.md §summary].

Every table row in the source files carries the exact command that produced it; they are not reproduced here [build/U20/RESULTS.md §1; build/h24/RESULTS.md §3; build/h30/RESULTS.md §0; build/U19/RESULTS.md §1].

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[MISSING] — open items this draft cannot fill from the sources

1. **Title.** TITLE: (Anton).

2. **Authors, affiliations, acknowledgements.** Excluded by instruction.

3. **Whether the router repairs the 39/448 adversarial losses — closed on the fixed contact path only.** §4.9: the router repairs 6 of 39 against 5 of 39 for a PGS-100 control. Under the fixed contact path the measured repairs are 6 of 39 (PGS) and 7 of 39 (ADMM), 1 regression each. The 12 of 39 from position correction was produced under the old contact cache and is not re-run, so it closes nothing [build/P2close/RESULTS.md §A.3; build/U113/RESULTS.md §“Torn och U23”; build/U81/RESULTS.md §4].
4. **Uncontended batched throughput on ANYmal C and Talos.** The 17 561 env-steps/s and 82 868 directions/s for ANYmal C are contended local measurements; only the A1 batch was re-run on a dedicated L4 [build/U19/RESULTS.md §summary; build/U20/RESULTS.md §1].
5. **A like-for-like batched comparison against Kamino [R1] and MJX [R6].** Neither has published steps/s in the source material, so there is nothing to compare against [HUNT.md §Mätt/H25].
6. **Hardware validation.** No force/torque sensor, no physical robot, anywhere in the campaign (§5.5).
7. **The third ρ branch (near-open).** Its threshold τ is unidentifiable on the 13-scene router set and requires the 4096-env population [HUNT.md §Mätt/H23b].
8. **GPU router at packing scale — the convergence claim is closed, one defect is open.** The earlier “no $\eta > 0$ packing converges through the PCG x-step” is refuted: PCG-16 with the Ruiz ρ gives pack50 390 sweeps / 9.19e-9, dem_step 480 / 9.80e-9 and pack800 360 / 9.93e-9. Open: the generated clamp in the GPU Ruiz branch uses [1, 1] and moves $\|G\|$ by up to 25.59 %, and the bil_A path on pack800 is UNVERIFIED after an 85 s timeout [build/A60/RESULTS.md §1, §5, §10, §U55; build/U33/RESULTS.md §summary].
9. **Closing the eigvalsh host channel.** ρ^* still differs by 111/33/0 ulp; no cheap fix found [HUNT.md §Mätt/fuse_fp-fix].
10. **Leave-one-out error for the generalized $n_c \leq 4$ router rule.** The rule was chosen on the same 21 scenes and its LOO was not reported [HUNT.md §Mätt/U33].
1. **GPU patch lumping for the forward path.** Built for the gradient path (rank reasons); the cone needs three more kernels and the forward path is launch-bound at these sizes [HUNT.md §Mätt/U2b].
2. **Licence and code-release terms for the solver.** Not stated in any source file.
3. **Figures.** No figure was produced by any source lane; all evidence is tabular.
4. **Gradient comparison against MJX and Brax autodiff.** Both differentiate through the solver iterations in JAX; no run of either exists in this campaign (§5.7).
5. **A gradient comparison against Dojo on one shared scene.** Dojo differentiates a relaxed intermediate problem by the implicit function theorem; no run of Dojo exists in this campaign (§5.7).
6. **Reverse-mode autodiff of the unrolled ADMM iterations (wp.Tape) on the same scene.** The solver is written in Warp and the tape is available; the comparison was not run (§2.3, §5.7).
7. **The determinism price against another deterministic reduction.** int64 fixed point is measured against float32 atomics only; segmented sort-based reduction, per-body fixed-order reduction and compensated summation were not measured (§3.4, §5.7).
8. **The packing and U23 penetration numbers under the fixed contact cache — closed for $N = 100$, $N = 300$ and the U23 set.** Reported in §4.7: both controls were re-run under the fixed cache (PGS 6 of 39, ADMM 7 of 39, 1 regression each), the tower gives 0 of 8 at $S = 4$ for both solvers, and the $N = 100$ and $N = 300$ piles were re-run from the initial state. $N = 1\,000$ is a four-substep probe from a checkpoint, not a run from the initial state and not a rest test [build/U113/RESULTS.md §“Torn och U23”, §“Högen från initialtillstånd”, §“Modal $N=1000$ ”].